



Pre installation

Installation

Operating

**LC 20MT Chilled Water System for
GE Signa *TwinSpeed* MRI Systems**

**Technical Publication 472
Revision 2**

The People of Ellis and Watts are committed to set goals and objectives to meet the expectations and needs of our customers. Responsibility and ownership at the point of operation will enhance the success of our mission

Ellis & Watts International, Inc.
Batavia, Ohio 45103

REVISION HISTORY PAGE	4
1 INTRODUCTION	5
1-1 CONTACT ELLIS & WATTS.....	5
1-2 WARRANTY REPLACEMENT PART PROCEDURE.....	5
1-3 INSTALLATION TOOLS & EQUIPMENT.....	6
1-3-1 INSTALLATION RESPONSIBILITY MATRIX.....	7
1-4 EXPLANATION OF NOTES, CAUTIONS, AND WARNINGS	8
1-5 SAFETY WARNINGS	9
2 SHIPPING AND DELIVERY DATA.....	12
2-1 SHIPMENT	12
2-2 STORAGE REQUIREMENTS	13
3 PRE-INSTALLATION REQUIREMENTS	14
3-1 EQUIPMENT PLACEMENT.....	14
3-2 SITE ENVIRONMENT	15
3-2-1 OPERATING SPECIFICATIONS.....	15
3-2-2 POWER REQUIREMENTS.....	15
3-3 EQUIPMENT CLEARANCE REQUIREMENTS	16
3-4 COMPONENT DIMENSIONS.....	16
3-5 MINIMUM DOOR/HALLWAY SIZES.....	18
3-6 DOORWAY THRESHOLD CONSIDERATION	20
3-7 FLOOR LOADING AND WEIGHTS	20
3-8 RAISED FLOOR CONSIDERATIONS.....	21
3-9 ANCHORING AND SEISMIC CONSIDERATIONS.....	22
4 UNCRATING AND INSTALLATION PROCEDURES.....	23
4-1 UNCRATING AND RECEIVING INSPECTION	23
4-1-1 TSCC.....	23
4-2 SHIP LOOSE ITEMS.....	28
5 EQUIPMENT INSTALLATION	30
5-1 INSTALLING THE VIBRATION ISAOLATORS (Optional – Used with Tie Down Brackets).....	30
5-2 OPTIONAL SEISMIC TIE DOWN BRACKETS.....	33
5-3 AIR DAM INSTALLATION.....	36
5-4 ELECTRICAL INSTALLATION.....	37
5-5 ASSEMBLING THE INTERCONNECTING HOSES.....	39
5-6 CRYO COOLER COMPRESSOR INSTALLATION.....	41
6 SYSTEM OPERATION.....	44
6-1 OPERATOR START-UP PROCEDURE.....	44
6-1-1 STARTING UP THE TSCC AND ASSOCIATED EQUIPMENT	44
6-1-2 SHUTTING DOWN THE TSCC.....	46
6-1-3 MAINTENANCE.....	46
6-1-4 GENERAL TROUBLESHOOTING.....	46
6-2 PLC OPERATION.....	47
6-2-1 OPERATOR INTERFACE DISPLAY DESCRIPTION:	47
6-2-2 BUTTON DESCRIPTION	47
6-2-3 MENU LISTS.....	48
6-2-4 CHANGING THE SET POINTS.....	48
TERMS & ACRONYMS.....	50
INDEX OF TABLES	51
INDEX OF ILLUSTRATIONS.....	52

<u>M</u>A<u>T</u>E<u>R</u>I<u>A</u>L <u>S</u>A<u>F</u>E<u>T</u>Y <u>D</u>A<u>T</u>A <u>S</u>H<u>E</u>E<u>T</u>S.....	53
FI DRAWINGS	60

REVISION HISTORY PAGE

REVISION NUMBER	DATE	REASON FOR CHANGE	AUTHOR
PRELIMINARY	5/23/01	NEW PRODUCT RELEASE	CAR
A	7/20/01	REVISIONS PER GE	CAR
B	8/14/01	REVISIONS PER GE	CAR
C	8/31/01	REVISIONS PER GE	CAR
0	9/19/01	INITIAL RELEASE	CAR
1	10/22/01	UPDATE TO PAGE 44	CAR
2	11/29/01	UPDATES TO PAGES 11, 60-65	CAR

1 INTRODUCTION

1-1 CONTACT ELLIS & WATTS

If a non-service issue arises concerning the equipment or the documentation provided with the chiller you can contact Ellis & Watts in writing at the following address, by phone, or via E-mail at the Ellis & Watts web page.

**Ellis & Watts International
Thermal Sales
4400 Glen Willow Lake Lane
Batavia, OH 45103**

Phone: 513-752-9000

Internet: WWW.ELLISWATTS.COM

Ellis & Watts provides 24-hour service support in the event that the Ellis and Watts equipment fails to perform properly. Service calls are processed more easily if you provide the service support department with the chiller **model number, serial number, the site name and location, and a description of the problem including the alarm or fault displayed on the operator interface**. The service support department will dispatch a qualified local service technician to correct the problem.

The 24 hour service support department phone numbers are:

Domestic Toll Free: 1-888-744-3195

International: 513-768-3195

1-2 WARRANTY REPLACEMENT PART PROCEDURE

To initiate a warranty claim with Ellis & Watts, contact the Thermal Sales Department. The equipment model number and serial number are required to process a warranty claim. The Thermal Sales representative will issue a Return Goods Authorization (RGA) number. The RGA number must be on the returned part's package.

NOTICE

Items returned without an RGA number will not be processed until the necessary information is obtained from the sender.

1-3 INSTALLATION TOOLS & EQUIPMENT

Item Description	QTY	Used For	Supplied By
Forklift, 2 ton (1814 kg) Minimum	1	Lifting and Moving System Cooling Cabinet and Associated Equipment	Installer
10 foot (3.05 m) Stepladder	2	Uncrating Equipment	Installer
Crowbar, 2 foot (.61m)	1	Uncrating Equipment	Installer
Pry bar, lever, 5000 lb. (2250kg)	1	Lifting Equipment	Installer
4 Conductor power cable	1	Power for System Cooling Cabinet	Installer
4X4X4 inch wood block (10.16X10.16X10.16 cm)	4	Supporting System Cooling Cabinet, Hose Assembling	Installer
Wood Block Approximately 1X4X6 inch (2.54 x 10.16 x 15.24cm)	1	Supporting System Cooling Cabinet	Installer
Metal Snips	1	Cut banding Straps	Installer
2 Eye, Lift Straps 2 inch W x 18 feet long (5.08cm W x 5.49m Lg.)	2	Lifting System Cooling Cabinet off Shipping Skid	Installer
C Clamp, 10 inch (25.4cm)	2	Securing Straps	Installer
Drill 3/8 inch minimum	1	Drill mounting holes	Installer
3/8 inch (M10) Hardware Length to be determined	6	For tie down brackets	Installer
Light oil or soapy water	1 oz	Lubricating Hose Barbs	Installer
LocTite 567 or equivalent	4 oz	Sealing Pipe Threads	Installer
Razor Knife	1	Cutting Hose	Installer

Table 1 **INSTALLATION TOOLS AND SUPPLIES**

1-3-1 Installation Responsibility Matrix

	Cust. Contr	GE Service	E&W Service	Who Pays
Once Cooling Cabinets Arrive				
Unload TSCC from Truck See Section 4-1	X			Customer
Move TSCC to Equipment Room and mount (Use seismic Tie down brackets and vibration isolators where required by law) See Section 3-8, 9	X			Customer
Connect customer supplied power cable from MDP to TSCC See Section 5-1	X			Customer
Once Magnet Arrives				
Install water lines from TSCC to 1) Gradient Coil 2) Cryo Cooler Compressor (Note: If Gradient is not connected, then it must be short looped on TSCC) See Section 5-3		X		GE
Fill TSCC with Propylene Glycol See Section 6-1		X		GE
Start up TSCC and ensure proper phasing and no coolant leaks See Section 6-1		X		GE
Verify proper operation per E&W Chiller Startup Check List after Cryo Cooler Compressor load is present See Section 6-1			X	E&W

Table 2 **INSTALLATION RESPONSIBILITY MATRIX**

1-4 EXPLANATION OF NOTES, CAUTIONS, AND WARNINGS

A **NOTE** is used to provide supplemental information or direction.



CAUTION

A **CAUTION** is used to call particular attention to a step of a procedure, which, if not strictly followed, could result in damage to, or destruction of equipment.



WARNING

A **WARNING** is used to call attention to a particular step of a procedure, which, if not strictly followed, could result in serious injury, or death of personnel.

1-5 SAFETY WARNINGS

WARNING

NO PERSON SHALL REACH INTO THE UNIT ENCLOSURE, WHICH CONTAINS LIVE ELECTRICAL CIRCUITS AS WELL AS ROTATING PARTS, FOR THE PURPOSE OF INSPECTING, SERVICING, ADJUSTING, OR MAINTENANCE EXCEPT IN THE PRESENCE OF SOMEONE CAPABLE OF RENDERING AID. CONTACTING SUCH ITEMS COULD CAUSE SERIOUS BODILY INJURY AND OR DEATH.

WARNING

PERSONNEL WORKING NEAR HIGH VOLTAGES SHOULD BE IN THE PRESENCE OF SOMEONE HAVING RESUSCITATION SKILLS. COMING IN CONTACT WITH HIGH VOLTAGE COULD CAUSE SERIOUS BODILY INJURY AND OR DEATH.

WARNING

TURN OFF AND TAG OUT THE MAIN POWER AT THE DISCONNECT, DURING ANY SERVICING OR MAINTENANCE OPERATIONS. REMOTE STARTING COMPONENTS OR DIRECT CONTACT WITH ELECTRICAL COMPONENTS MAY RESULT IN INJURY TO PERSONNEL.

WARNING

TURN OFF AND TAG OUT THE DISCONNECT TO THE UNIT BEFORE WORKING ON CONTACTORS OR OTHER COMPONENTS IN THE ELECTRIC BOX. WHEN THE MAIN POWER SUPPLY TO THE UNIT IS ON, THE LIVE SIDE OF EACH CONTACTOR CONTAINS HIGH VOLTAGE REGARDLESS OF ANY OTHER SWITCH SETTING. DO NOT ATTEMPT MAINTENANCE PROCEDURES INSIDE EQUIPMENT WITH HIGH VOLTAGE ON. REMOVE POWER AND TAG OUT THE DISCONNECT. GROUND ELECTRICAL COMPONENTS PRIOR TO TOUCHING TO PREVENT BODILY INJURY.

WARNING

NO SMOKING OR OPEN FLAMES WITHIN 50 feet (15 M) OF REFRIGERANT SYSTEM. REFRIGERANT COMING IN CONTACT WITH LIT CIGARETTES CAN CREATE AN IRRITATING AND POSSIBLY TOXIC GAS.

SAFETY WARNINGS CONTINUED

WARNING

STAND CLEAR DURING LIFTING OPERATIONS. OPERATOR ERROR AND/OR EQUIPMENT MALFUNCTION CAN CREATE OPPORTUNITIES FOR PERSONAL INJURY.

WARNING

THE UNIT HAS A CLOSED LOOP HIGH PRESSURE REFRIGERANT SYSTEM. DO NOT CLOSE, OPEN OR ADJUST VALVES IN THE REFRIGERANT SYSTEM UNLESS A SPECIFIC INSTRUCTION STATES THAT VALVES MUST BE OPENED, CLOSED OR ADJUSTED. PERSONNEL COULD BE INJURED OR EQUIPMENT COULD BE DAMAGED.

WARNING

BEFORE BRAZING OR SOLDERING ANY REFRIGERATION COMPONENT, RECOVER REFRIGERANT PER US ENVIRONMENTAL PROTECTION AGENCY (EPA) APPROVED PROCEDURE TO RELEASE THE PRESSURE FROM THE REFRIGERATION SYSTEM. SUDDEN DECOMPRESSION OF REFRIGERANT PIPING CAN CAUSE SERIOUS PERSONAL INJURY.

SAFETY WARNINGS CONTINUED**WARNING**

USE EXTREME CARE WHEN BRAZING OR SOLDERING IN THE PRESENCE OF R-134a. WHEN R-134a COMES IN CONTACT WITH HOT METAL, AN IRRITATING GAS CAN FORM. HAVE SUFFICIENT FRESH AIR AND VENTILATION WHEN BRAZING, SOLDERING OR USING A HALIDE TORCH LEAK DETECTOR.

WARNING

THE COMPRESSOR MUST ALWAYS BE OPERATED WITH THE COMPRESSOR DISCHARGE SERVICE VALVE IN THE BACK-SEAT (FULLY OPEN) POSITION. FAILURE TO DO SO CAN CAUSE SERIOUS COMPONENT FAILURE.

WARNING

WHEN AN ABNORMAL CONDITION IS NOTED AND PROCEDURES IN THIS MANUAL DO NOT SPECIFICALLY RELATE TO THE NOTED CONDITION, WORK SHALL BE STOPPED. TECHNICALLY QUALIFIED PERSONNEL SHALL GIVE GUIDANCE ON THE ABNORMAL CONDITION BEFORE WORK CAN BE CONTINUED, TO PREVENT POSSIBLE PERSONAL INJURY OR EQUIPMENT DAMAGE.

2 SHIPPING AND DELIVERY DATA

The General Electric Medical Systems (GEMS) part number 2281346-3 High Voltage Indoor Air Cooled Unit (400V – 50 Hz , 480V – 60 Hz) is the Ellis and Watts chiller model LC20MT. The LC20MT consists of one cabinet which houses the liquid chiller known as the Twin System Cooling Cabinet (TSCC).

2-1 SHIPMENT

For domestic shipment, the TSCC will be shipped to the construction site. International shipment shall be shipped to a collection point where all components for the magnet will be collected and shipped to the construction site.

Unit	Height	Width	Depth	Weights
TSCC	80 inches (203.2 cm)	68 inches (172.72 cm)	36 inches (91.44 cm)	1550 (703 kg)
Ship Loose Box		28.5 inches (72.5 cm)	58.8 inches (149.4 cm)	150 lbs (68 kg)

Table 3 **SHIPPING DIMENSIONS**

The ship loose items are boxed and shipped together on a wooden skid. Domestic shipments are covered with cardboard and banded on wooden skids. International shipments are domestic packages shipped in crates.

2-2 STORAGE REQUIREMENTS

The non-operation environmental conditions are as follows:

ITEM	REQUIREMENT
Ambient Temperature	-40°F (-40°C) to 158°F (70°C)
Humidity	5% to 100% RH
Altitude	984 feet (300 m) below sea level 17,060 feet (5,200 m) above sea level
Acceleration	+/- 1.5 G any axis
Magnetic Field	30 Gauss
NOTE: Do not expose to water (rain, snow)	

Table 4 **ENVIRONMENTAL CONDITIONS**

NOTE:

The TSCC should be kept in its shipping crate until it has been moved to the installation site. When uncrating, care should be taken to prevent damage.

3 PRE-INSTALLATION REQUIREMENTS

3-1 EQUIPMENT PLACEMENT

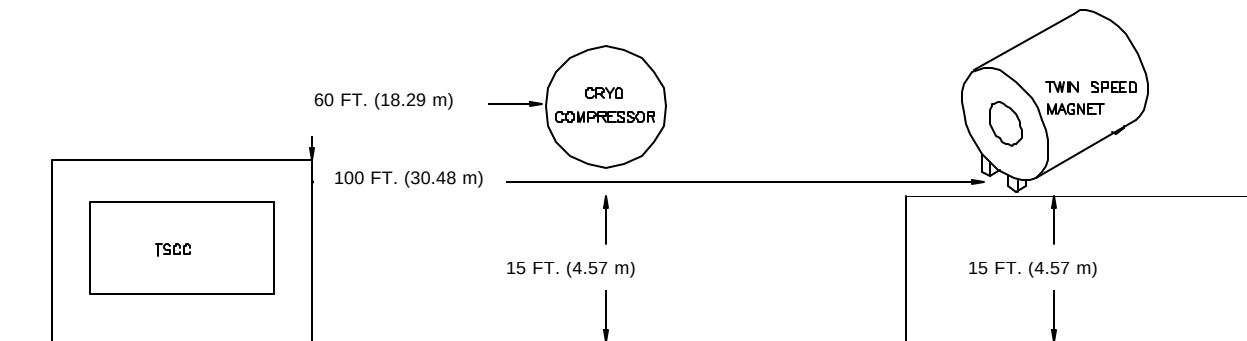


Illustration 1 **PLACEMENT OF TSCC TO MAGNET**

- The TSCC and the Gradient Coil located inside the magnet must not be separated by a distance greater than 100 feet (30.48 m) water line length.
- The TSCC and the Shield/Cryo Cooler Compressor Cabinet must not be separated by a distance greater than 60 feet (18.29 m) water line length.
- The TSCC vertical separation from the Gradient Coil located inside the magnet and the Cryo Cooler Compressor must not exceed 15 feet (4.57 m) above or below the Gradient Coil and Cryo Cooler Compressor.

3-2 SITE ENVIRONMENT

3-2-1 Operating Specifications

ITEM	REQUIREMENT
Ambient Temperature	5 ⁰ C to 40 ⁰ C
Humidity	10% to 90%
Altitude	984 feet (300 m) below sea level to 11,800 feet (3,597 m) above sea level
Noise Levels	75 dBa @ 1 meter in all directions
Magnetic Field	30 Gauss

Table 5 **OPERATING SPECIFICATIONS**

3-2-2 Power Requirements

ITEM	REQUIREMENT
Wire - <u>NOTE</u>	2AWG (6.59 mm) maximum 6 AWG (4.11 mm) minimum 3 conductor w/ground
Power Consumption	13 KVA (maximum continuous)
Voltage Requirements	480 V +/- 10% 60 Hz +/- 3 Hz 400 V +/- 10% 50 Hz +/- 3 Hz
Circuit Size	50 AMP (maximum)
<u>NOTE:</u> The TSCC will be hardwired to the MDP (Main Distribution Panel) with a four conductor power cable supplied by the Installer. Size to be determined by the Installer.	

Table 6 **POWER REQUIREMENTS**

3-3 EQUIPMENT CLEARANCE REQUIREMENTS

It is necessary to maintain proper distances at the top, front and back of the TSCC cabinet in order to have a good service area. There is no requirement for clearance at the sides of the cabinet, but any allowances would aid in the service of the TSCC cabinet.

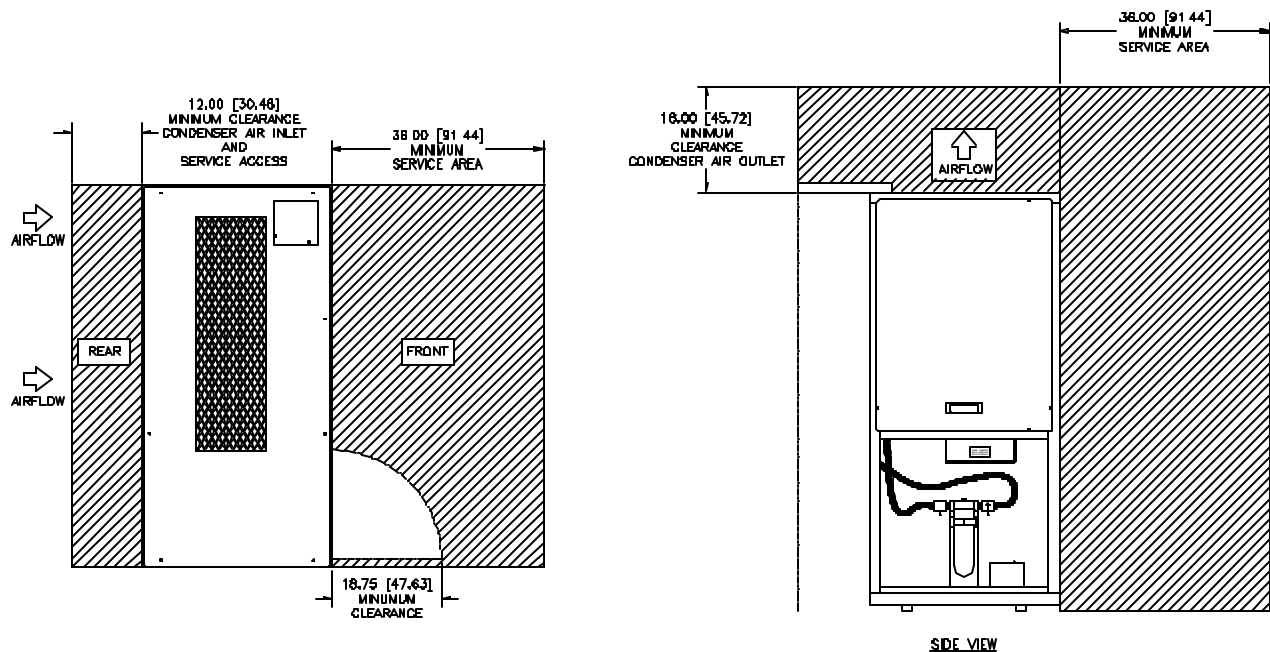


Illustration 2 NECESSARY CLEARANCES FOR TSCC

3-4 COMPONENT DIMENSIONS

Uncrated Dimensions

Component	Height	Width	Depth	Installed Wet Weight
TSCC	71.26 inches (181 cm)	65 inches (165.10 cm)	32.5 inches (82.55 cm)	1410 pounds (640 kg)

Table 7 UNCRATED DIMENSIONS

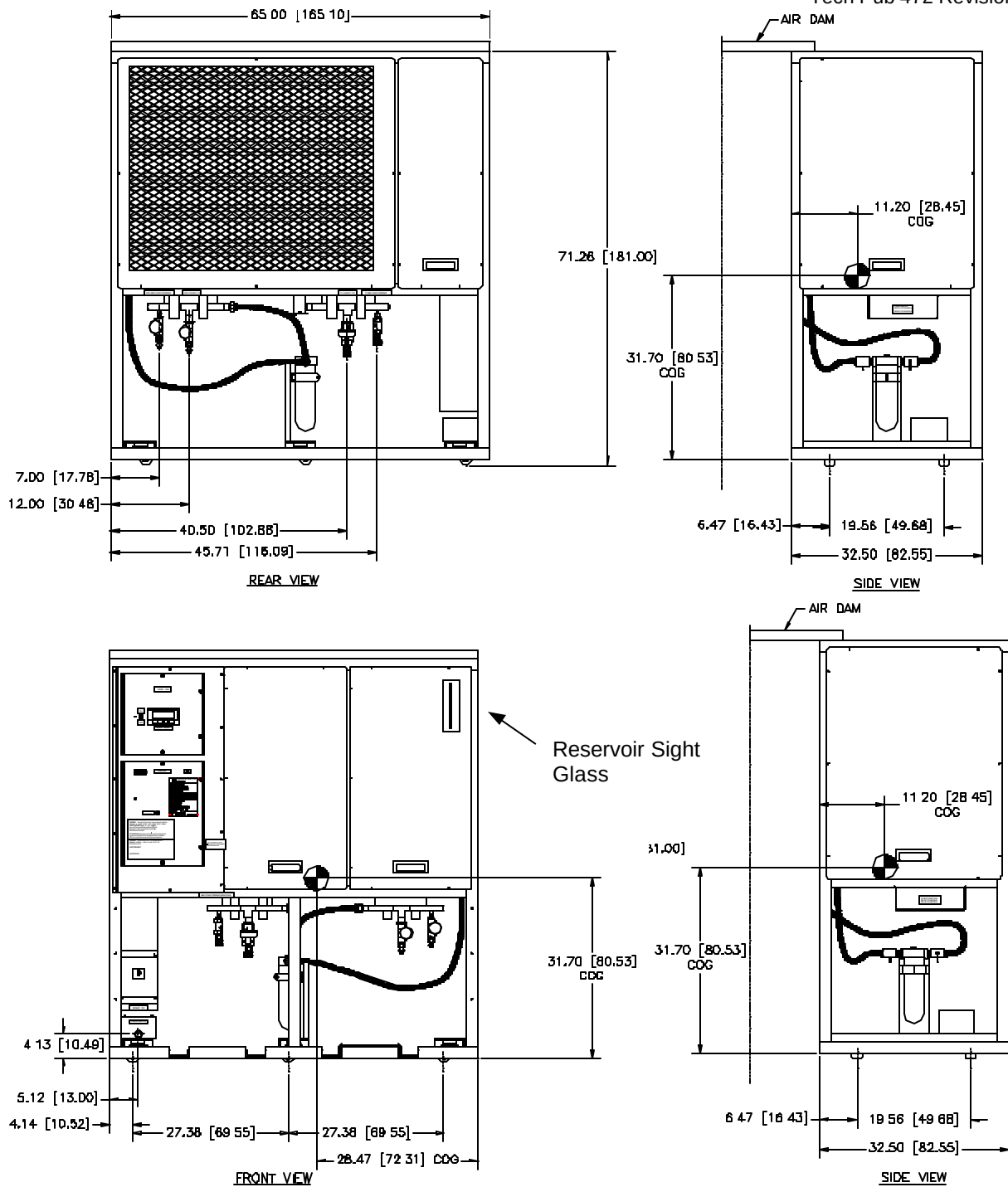
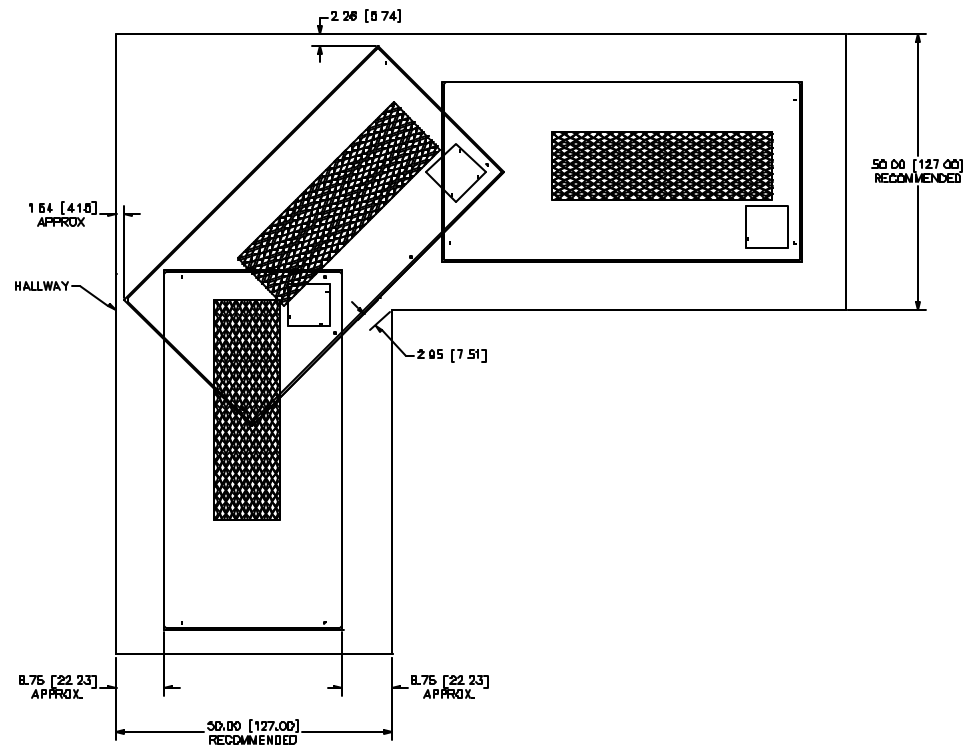


Illustration 3 TSCC OVERALL DIMENSIONS

3-5 MINIMUM DOOR/HALLWAY SIZES

Component	Hallway / Door Height	Hallway Width Note	Doorway Width
TSCC	76 inches 193.04 cm	50 inches 127 cm	50 inches 127 cm
NOTE: Minimum width for 90° turns			

Table 8 **DOOR/HALLWAY SIZES**Illustration 4 **MINIMUM HALLWAY DIMENSION FOR TURNS ([] = cm)**

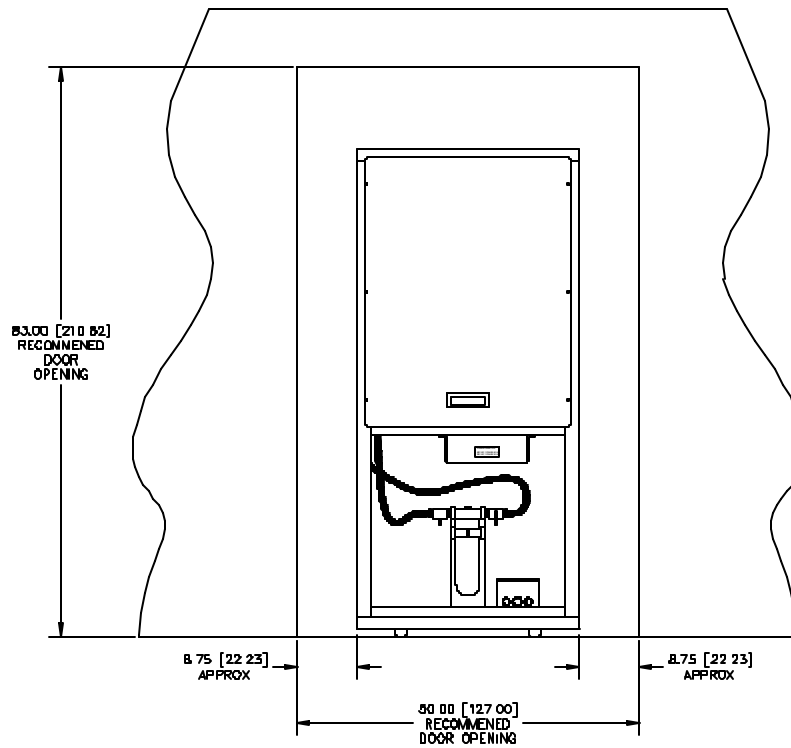


Illustration 5 **MINIMUM DOORWAY DIMENSIONS** ([] = cm)

3-6 DOORWAY THRESHOLD CONSIDERATION

The clearance from the bottom of the TSCC cabinet to the floor is approximately 0.89 inches (2.26 cm) therefore doorway threshold should be as smooth as possible or special considerations have to be made to move cabinet over a threshold height of 0.5 inches (1.27 cm)

3-7 FLOOR LOADING AND WEIGHTS

EQUIPMENT	INSTALLED WEIGHT
TSCC	1410 lbs. (640 Kg)

Table 9 **FLOOR LOADING REQUIREMENTS**

3-8 RAISED FLOOR CONSIDERATIONS

The TSCC cabinet sits on six pneumatic vibration isolators to reduce vibrations transmitted into the floor, which can affect image quality of the MRI equipment. These vibration isolators are required when using the tie down brackets. The placement of these isolators need to be taken into consideration when laying out the raised floor framing in order to support the footprint weight per isolator, i.e., 295 Lbs. (133.8 Kg) per isolator.

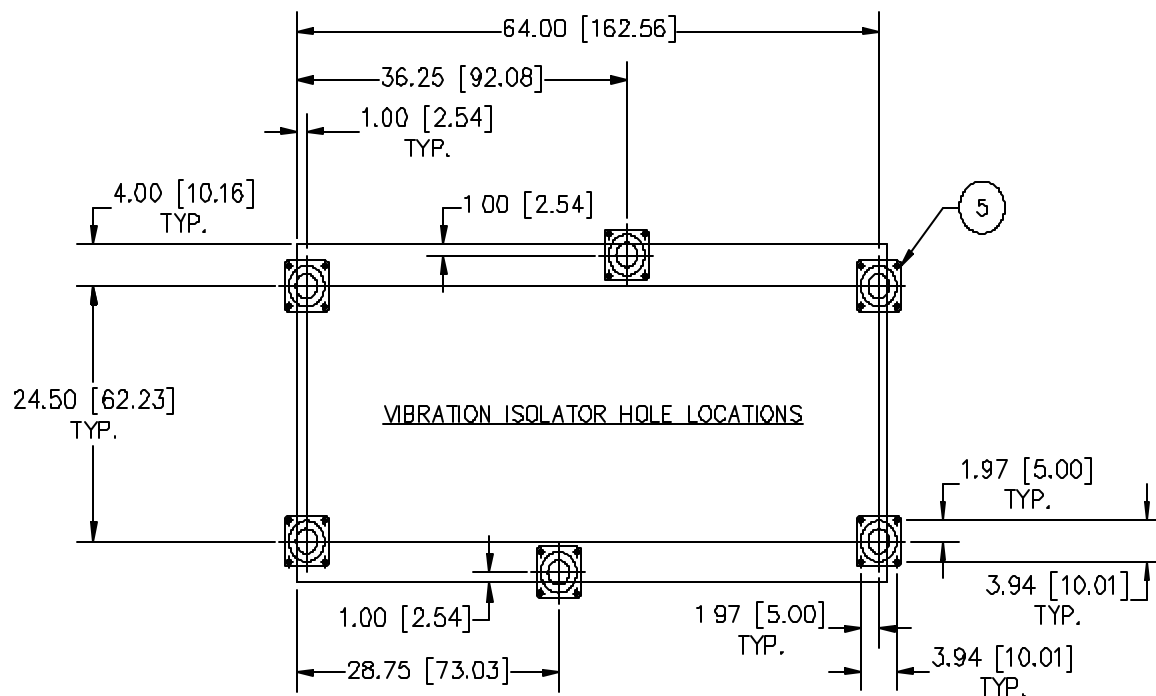


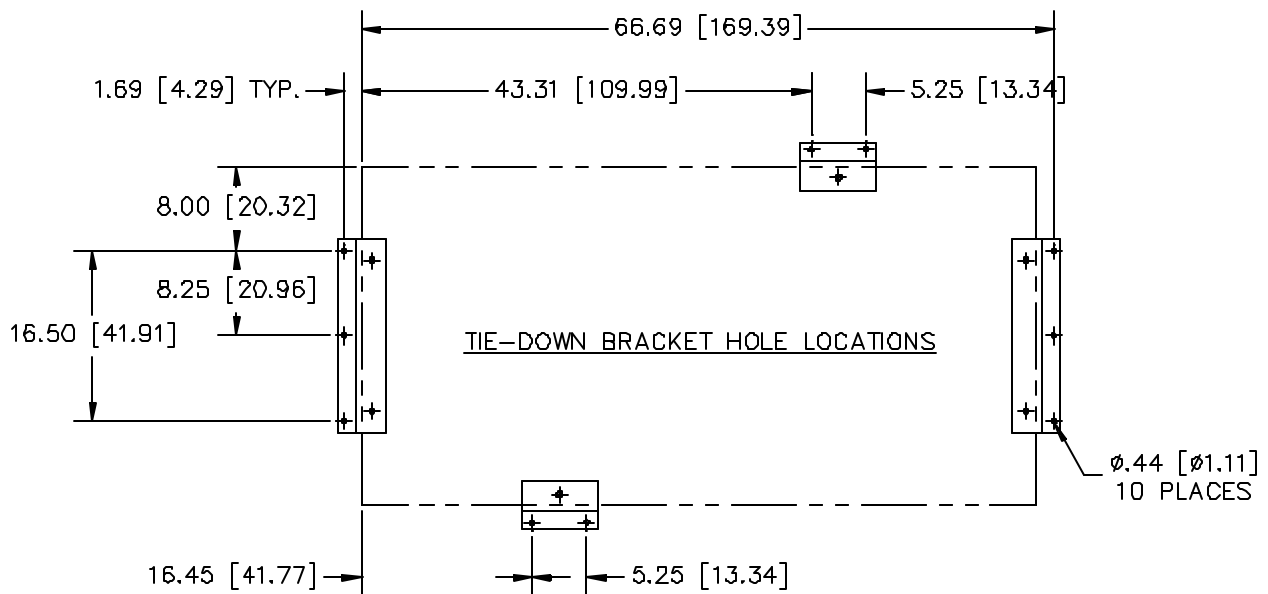
Illustration 6 **VIBRATION ISOLATOR LAYOUT DIMENSIONS**

Refer to table 10, item number 5

3-9 ANCHORING AND SEISMIC CONSIDERATIONS

The TSCC cabinet has tie down brackets included in the ship loose parts, which should be used to aid in preventing the cabinet from shifting during a seismic event. Site Architects and Engineers need to take into consideration the raised floor-framing layout in order to have substantial support for the tie down brackets.

Sites requiring seismic anchoring by code should have the site architect and engineer review the response spectra and/or United Builders Code (UBC) for their location and then contact Ellis and Watts International for assistance in seismic planning of the site.



NOTE: USE SEISMIC TIE-DOWN BRACKETS WHERE APPLICABLE.

Illustration 7 TSCC TIE DOWN BRACKET PLACEMENT

4 UNCRATING AND INSTALLATION PROCEDURES

4-1 UNCRATING AND RECEIVING INSPECTION

NOTE

Items 1 through 3 are for International shipments only.

4-1-1 TSCC

1. Using a #2 Phillips screwdriver, unscrew the wood screws securing the top of the crate. Refer to Section 2-1 for dimensions.

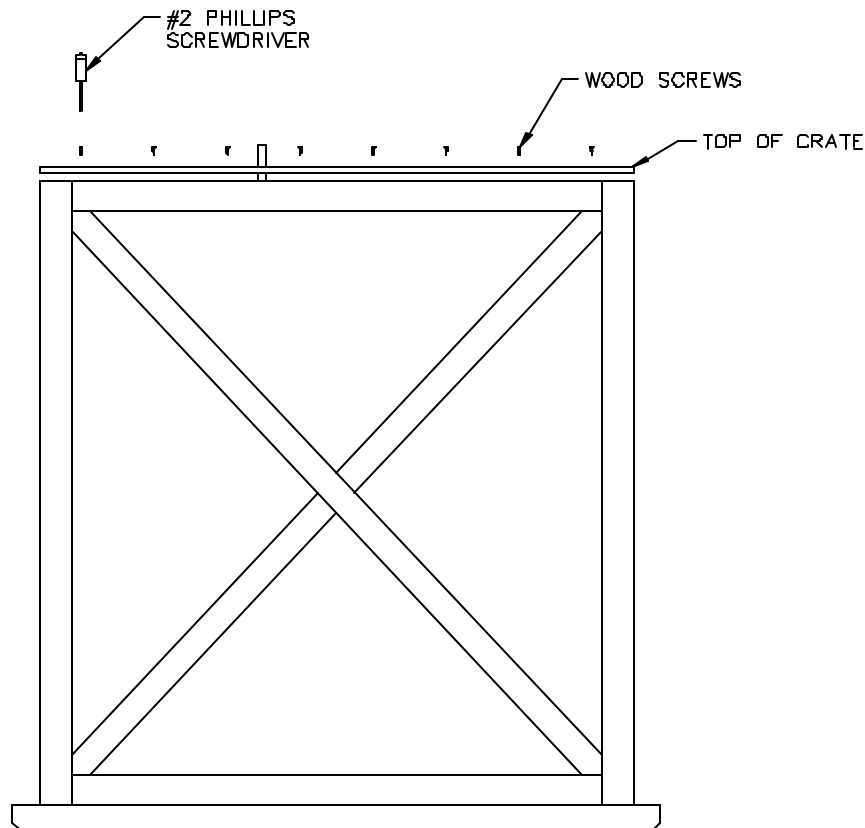


Illustration 8 **INTERNATIONAL CRATE EXAMPLE**

2. Remove top of the crate and place in an area clear of the uncrating area.
3. Use a pry bar to carefully remove the front, sides, and back of the crate and place in an area clear of the uncrating area.

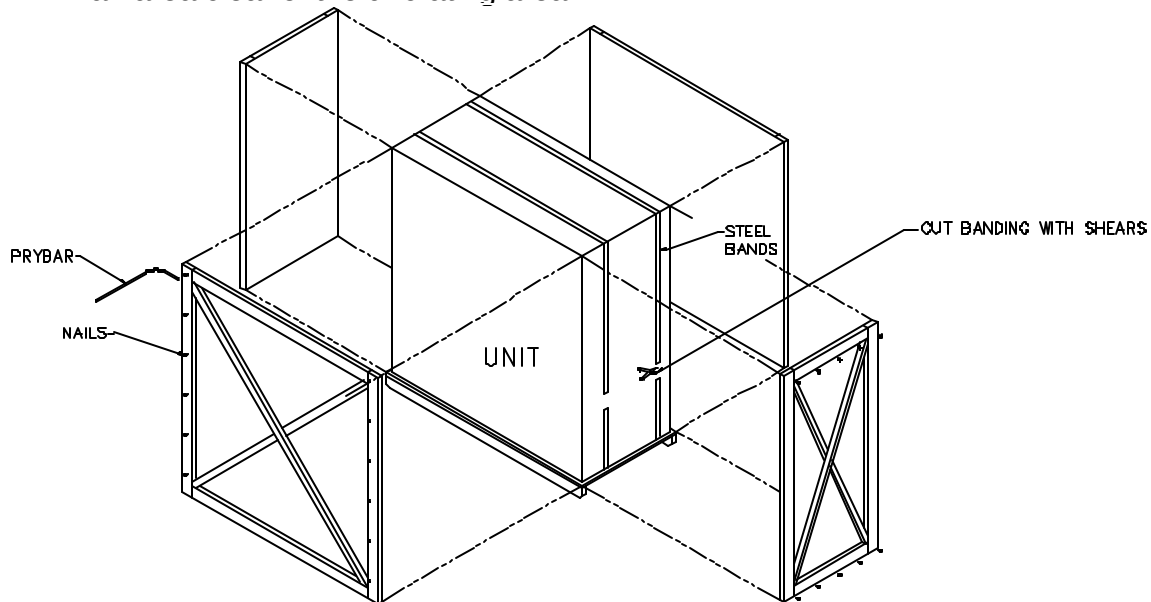


Illustration 9 **UNCRATING**

CAUTION

Banding straps can be under high tension and can whip out when cut.

4. Remove banding by cutting with metal cutting hand shears.
5. CAREFULLY remove plastic.
6. Inspect equipment for any damage. Contact Ellis and Watts and the shipping company as soon as possible and report any discovered damage.
7. Remove the ship loose cardboard boxes from the crate and place in secure location for use during installation.

NOTE: These boxes contain the ship loose items and must be stored in a safe location for later use. Boxes can be inventoried using the list in section 4-2.

8. Using two eye and eye lifting straps approximately 2 inches (5.08 cm) wide and 18 feet (5.49 m) long, create a basket lift by pushing one end under unit (between board slats from front to back), and up opposite side until lift strap eyes are even.

9. Do this in two places, evenly spaced from, and as near as possible to sides of TSCC as fork spread will allow.
10. Using two 10 foot (3.05 m) stepladders on each end of the unit, hold up straps while bringing in forklift as near as possible to front or back of TSCC cabinet so lift forks can be inserted into strap eyes.

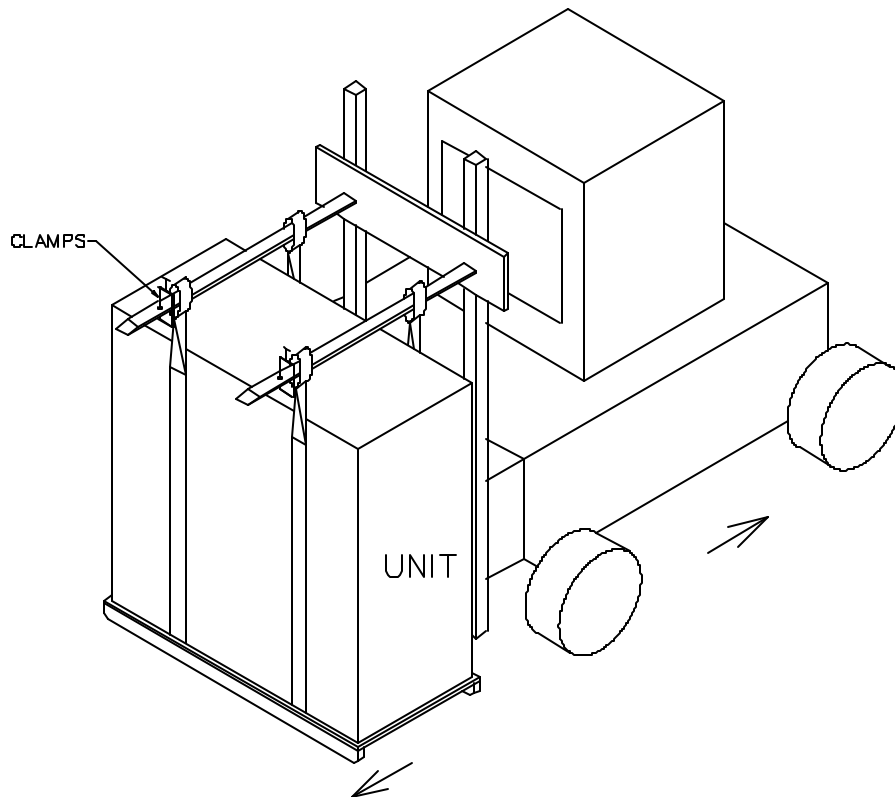


Illustration 10 TSCC LIFTING METHOD

11. After sliding lift forks through strap eyes, attach a C-clamp or similar clamping device, onto the end of the forks to prevent straps from sliding off the forks.

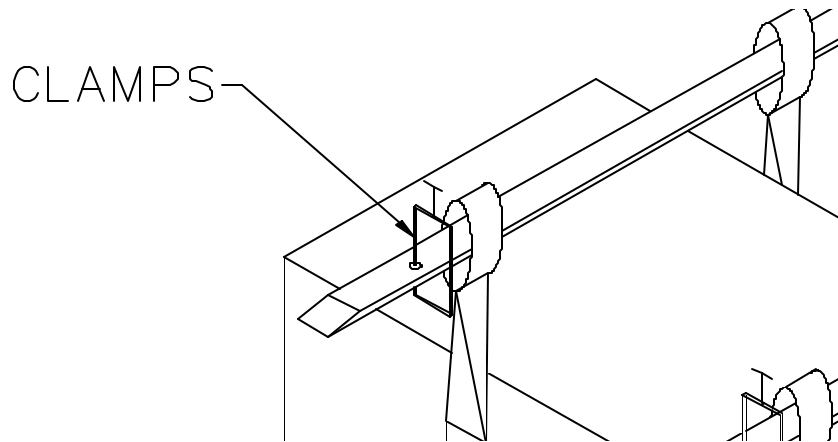


Illustration 11 C-CLAMP PLACEMENT

12. Gently lift TSCC cabinet until clear of shipping skid.
13. Remove skid, carefully move unit to smooth level area where TSCC cabinet can be moved on attached casters.
14. Remove slings from lifting forks and slide out from under TSCC cabinet.

The previous steps define the preferred method of removing the TSCC cabinet from the skid. Other methods of removing the TSCC cabinet from the shipping skid may be acceptable but could involve many more detailed steps. It is up to the judgment of the riggers and movers to determine the specific method of handling the TSCC cabinet.

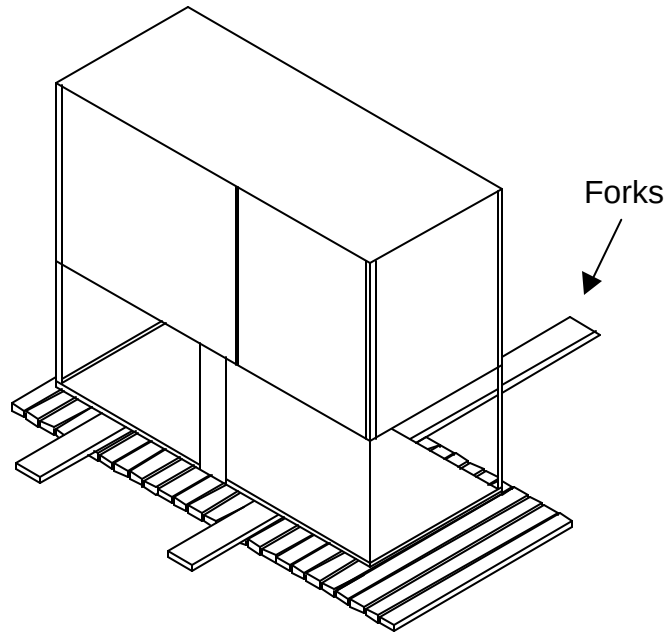


Illustration 12 **ACCEPTABLE ALTERNATIVE LIFTING METHOD**

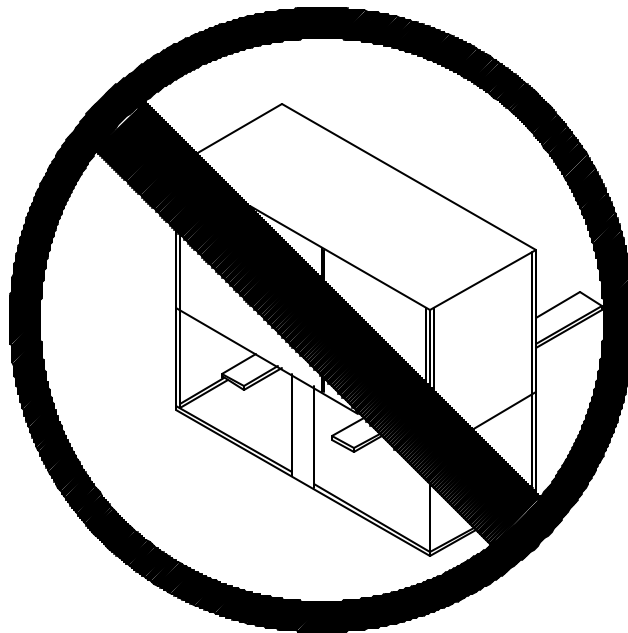


Illustration 13 **UNACCEPTABLE ALTERNATIVE LIFTING METHOD**

4-2 SHIP LOOSE ITEMS

The following tables list the ship loose items that are necessary for the proper installation of the Ellis and Watts equipment. These items were shipped inside the TSCC. The following tables can be used to verify the inventory of the shipped loose items.

The TSCC is shipped with the following field installation kits.

No.	Part No.	Ship Loose Items Description	QTY EA
1	T2625-0201-02	Seismic Tie Down Bracket	6
2	50-19X-616	Bolt, Hex Head, ½-13 by 4 Length SST	6
3	50-2Z-181	Nut, Lock, ½-13, SST	12
4	50-3E-38	Washer, Flat, ½" SST	18
5	T2625-282	Isolator, Vibration	6
6	01101	Schraeder/Presta Adapter	1
7	FI-0003	Drawing, Field Installation, Seismic Tie-Down Bracket	1

Table 10 **SEISMIC TIE DOWN KIT**

No.	Part No.	Ship Loose Items Description	QTY EA
1	T2590-111ZS	½ inch Q.D. Zinc Plated Steel Nipple, ½ inch NPT Female	1
2	J1476-213ZS	½ inch Hose Barb, ½ inch NPT Male	2
3	SS1045-08	Tube, ½ inch, Push-Lok, 120 Feet	1
4	T2637-140ZS	¾ inch Q.D. Zinc Plated Steel Nipple, ¾ inch NPT Female	1
5	J1476-206ZS	¾ inch Hose Barb, ¾ inch NPT Male	2
6	62-1-66	½ inch Hose Clamp	2
7	T2637-143ZS	¾ inch QD Coupler, ¾ inch NPT Female	1
8	T2637-113ZS	½ inch QD Coupler, ½ inch NPT Female	1
9	FI-0035	Drawing, Field Installation, Piping/Hose Connections	1

Table 11 **PIPING/HOSING CONNECTIONS KIT**

NO.	PART NO.	SHIPS LOOSE ITEMS DISCRIPTION	QTY EA
1	FI-0036	Drawing, Field Installation, Electrical, Power Communication Connections	1

Table 12 **OTHER DRAWINGS**

NO.	PART NO.	SHIPS LOOSE ITEMS DISCRIPTION	QTY EA
1	FI-0094	Air Dam	1
2	FI-0094	Drawing, Field Installation, Air Dam	1

Table 13 **AIR DAM**

NO.	PART NO.	SHIPS LOOSE ITEMS DISCRIPTION	QTY EA
1	T2625-279A	Filter, 10" Cartridge (Polypropylene)	1
2	PT1165-100	Fuse, Time Delay, 1.0 Amp, 500V	2
3	SS627-7-A20	Fuse, Time Delay, 20 Amp, 500V	6
4	GE P/N 2297672 E & W P/N J1520-5050	Propylene Glycol, per gallon	12 gal
5	FI-0018	Drawing, Field Installation, Spare Parts	1

Table 14 **SPARE PARTS**

NO.	PART NO.	SHIPS LOOSE ITEMS DISCRIPTION	QTY EA
1	T2625-0201-06	Cryo Cooler Ramp	2
2	45-1-33	Strap Assembly	2
3	FI-0033	Drawing, Field Installation, Air Dam	1

Table 15 **SHIELD COOLER INSTALLATION**

5 EQUIPMENT INSTALLATION

5-1 INSTALLING THE VIBRATION ISAOLATORS (Optional – Used with Tie Down Brackets)

The following instructions and illustrations are specifically for installation of the TSCC cabinet.

Refer to drawing FI-0003 for detailed illustrations.

1. Verify the following parts and quantities from the Seismic Tie-down Bracket/Vibration Isolator Kit, refer to Table10.
2. Locate area where TSCC cabinet is to be located. Ensure that the area meets the requirements of service clearance area, floor loading, vibration isolator placement considerations, and tie down bracket considerations.
 - 12 inches (30.48 cm) clearance at the back of the TSCC is required for the air dam installation.
 - Minimum of 36 inches (91.44 cm) clearance for the front of the TSCC
 - Minimum of 18 inches (45.72 cm) clearance at the top of the TSCC
3. Using the dimensions for the vibration isolator layout, carefully mark the areas where the isolators are to be mounted. Locate the Isolators in the ship loose items boxes, and place them in the marked off positions. Verify the dimensions to the print and to the lower frame of the TSCC (see Illustration 14).

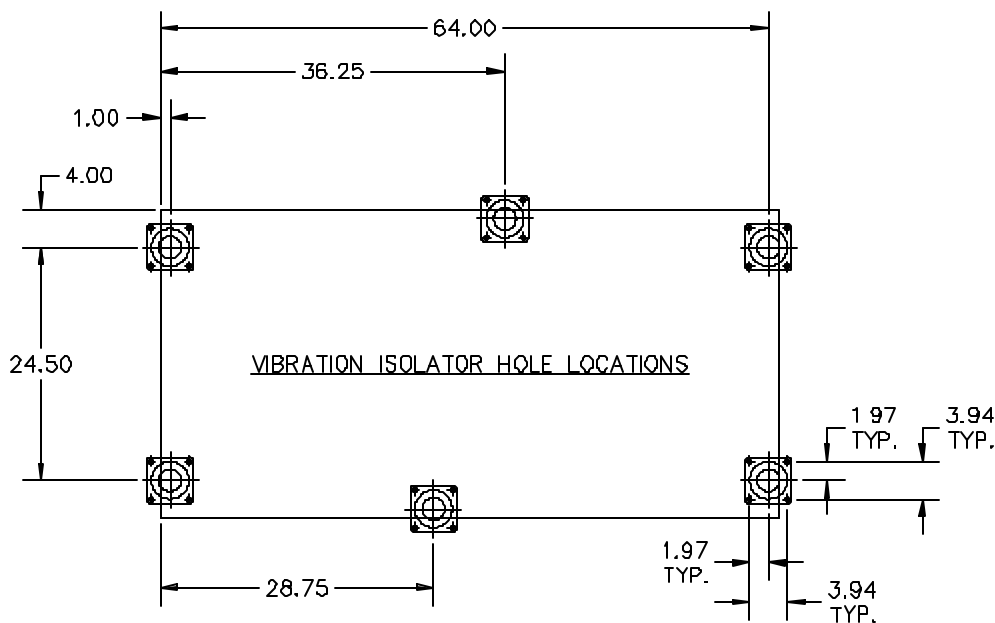


Illustration 14 **ISOLATOR LAYOUT PATTERN**

4. Once dimensions have been verified, mark isolator base mounting hole locations and drill out marked holes appropriate for installer supplied hardware that is appropriate for the site selected. Ellis and Watts recommends 3/8 inch (M10) Stainless Steel hardware.
5. Install floor anchors. The floor mounting hardware is not supplied due to variations in flooring materials
6. Roll the TSCC to the selected installation site.
7. Release any existing pressure inside the vibration isolator by pushing the valve stem with a pointed object such as a pen or a small screwdriver.
8. Raise the TSCC to a height of approximately 4 inches (10.16 cm) with a pry bar or equivalent, support the unit with temporary blocks.
9. Slide the isolator under the TSCC base ensuring the isolator valve stem is accessible.
10. Attach the isolator under the TSCC base using the 12 mm bolt, washer and nut supplied with the isolator. Thread the nut onto the bolt first, then the washer. Thread the nut up to top of bolt. Thread hardware through TSCC frame into top of isolator. Insert isolator base mounting hardware.

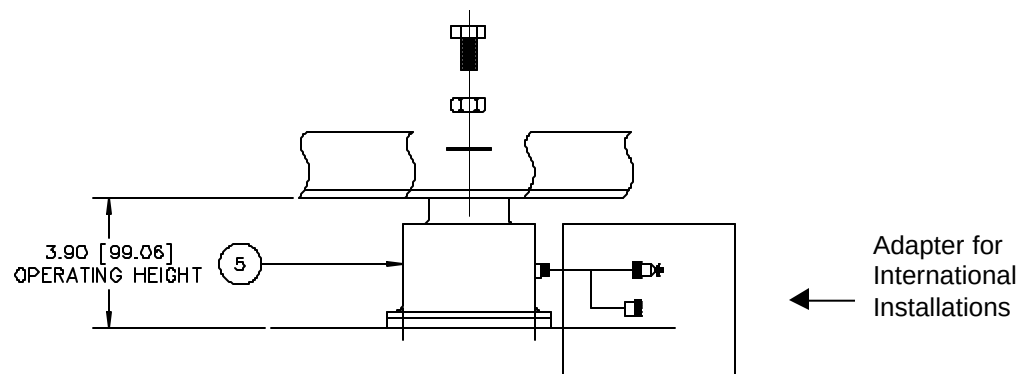


Illustration 15 VIBRATION ISOLATOR MOUNTING

Refer to table 10, item number 5

11. Determine torque requirements for isolator base mounting hardware and tighten to proper torque rating taking into account floor panel restrictions.

12. Tighten 12 mm bolt threaded through TSCC frame into vibration isolator (approximate 7/8 inch (2.22 cm)) by hand until tight; then turn with wrench another ½ turn.
13. Repeat previous steps for the other 5 isolators.

CAUTION

DO NOT EXCEED 60 psig (413.7 KPa), THE VIBRATION ISOLATOR'S MAXIMUM PRESSURE.

14. Inflate isolators evenly until TSCC weight is supported and temporary blocks can be removed.

CAUTION

Do not remove weight from vibration isolators before deflating. Isolators could over inflate and rupture.

15. Remove temporary blocks.
16. Bleed or add air to the vibration isolators to obtain a level operating height of 3.8 inches +/- .2 inches (9.65 cm ± .5 cm) (see Illustration 15).
17. Tighten nut on 12mm bolt until tight, tighten again with appropriate wrench until snug, then tighten ½ turn more.

5-2 OPTIONAL SEISMIC TIE DOWN BRACKETS

Refer to drawing FI-0003 that comes with the ship loose parts, for detailed illustrations.

1. Locate-Tie Down Brackets in the ship loose kit FI 0003
2. Locate positions to mount brackets per Tie-Down Brackets Layout pattern (see Illustration 16).

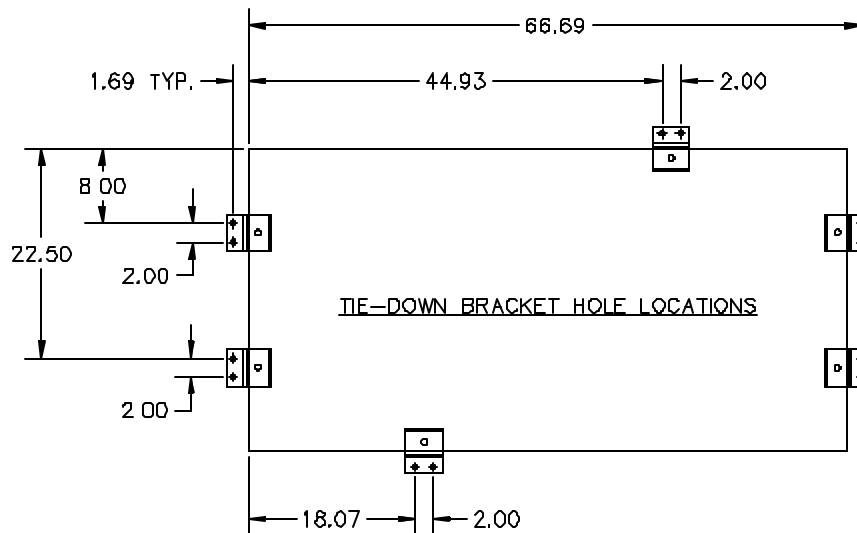
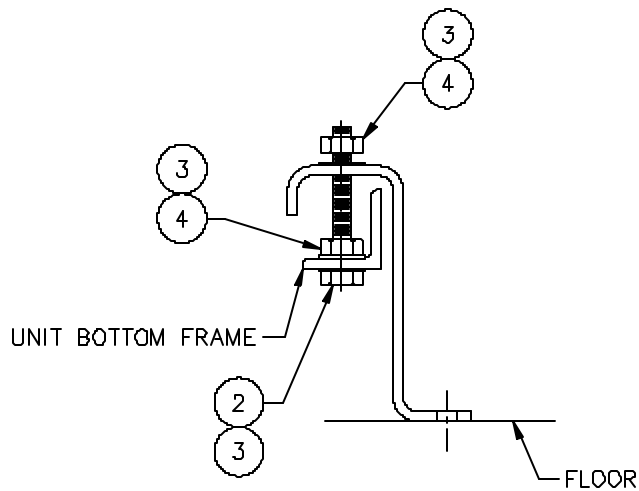


Illustration 16 **TIE-DOWN BRACKET LAYOUT PATTERN**

3. Locate mounting hardware in ship loose kit FI-0003. Refer to table 11, item number 2 to 4.
4. Taking one $\frac{1}{2}$ x 4 inch long bolt, slide one $\frac{1}{2}$ inch washer over bolt and then push the bolt through the hole in the base frame from the bottom up. Slide one $\frac{1}{2}$ inch washer over bolt and thread $\frac{1}{2}$ inch lock nut down until it is snug against the frame member and tighten approximately $\frac{1}{4}$ to $\frac{1}{2}$ turn more.

**Illustration 17 1/2 Inch Bolt Assembly**

Refer to table 10, item numbers 2 to 4

5. Place the tie down bracket over the 1/2 inch bolt (six per unit) at the bottom of the frame. Using the tie down bracket as a template, mark the location for the floor mounting hardware.
6. Repeat Steps 3-5 for the other 5 brackets.
7. Remove brackets and drill marked locations to the appropriate size for installer supplied hardware. Ellis and Watts recommend using 3/8 inch hardware.
8. Place the tie down bracket over the 1/2 inch bolt at the bottom of the frame. Locate one 1/2 inch lock-nut and one 1/2 inch washer. Slide washer down the bolt onto the tie-down bracket and thread the lock-nut down until it stops and then tighten approximately one more turn.

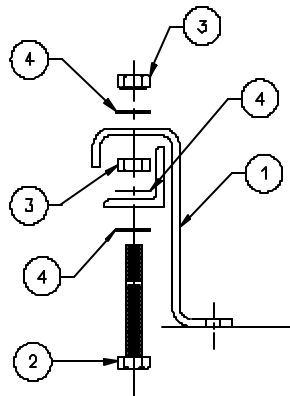


Illustration 18 **TIE-DOWN BRACKET MOUNTING**

Refer to table 10, items 1 to 4

9. Determine torque requirements for the floor mounting hardware and tighten to proper torque rating, taking into account floor panel restrictions.

5-3 AIR DAM INSTALLATION

Refer to drawing FI-0094 that comes with the ship loose parts, for detailed instructions.

1. Remove the four #10-32 phillips head screws located at the rear of the top access panel. See Illustration 19.
2. Place the air dam on top of the chiller. Align the mounting holes in the air dam with the access panel mounting holes.
3. Attach the air dam with the four #10-32 phillips head screws removed in step one.
4. With the chiller properly positioned, the rear of the air dam will be flush with the wall.

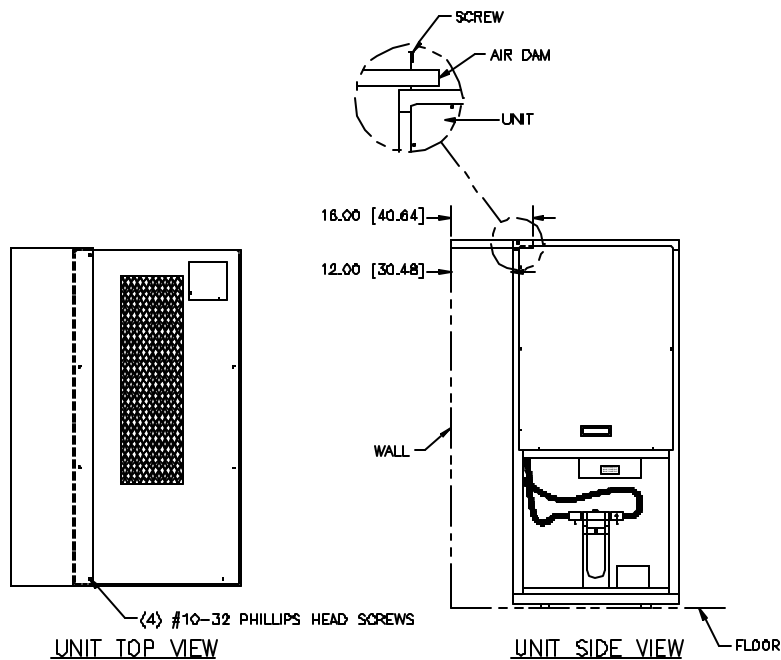


Illustration 19 AIR DAM INSTALLATION

5-4 ELECTRICAL INSTALLATION

Power Wiring

Connecting power to the TSCC should only be performed by trained electrical installation personnel that have a good knowledge of lock-out/tag-out procedures. Field wiring must be accomplished in accordance to the National Electrical code, ANSI/DEPA No. 70-1990 and or the Canadian Electrical Code, Part 1, CSA Standard C22.1-1990 and local laws and regulations as they exist.

Power Requirement

480 V +/- 10%, 60 Hz +/- 3 Hz, 3 phase, WYE or Delta
or
400 V +/- 10%, 50 Hz +/- 3 Hz, 3 Phase, WYE or Delta
Full Load current is 22.7 kVa.

Refer to drawing FI-0026A that comes with the ship loose parts, for detailed illustrations.

1. Locate the power junction box on the lower left side of the TSCC.

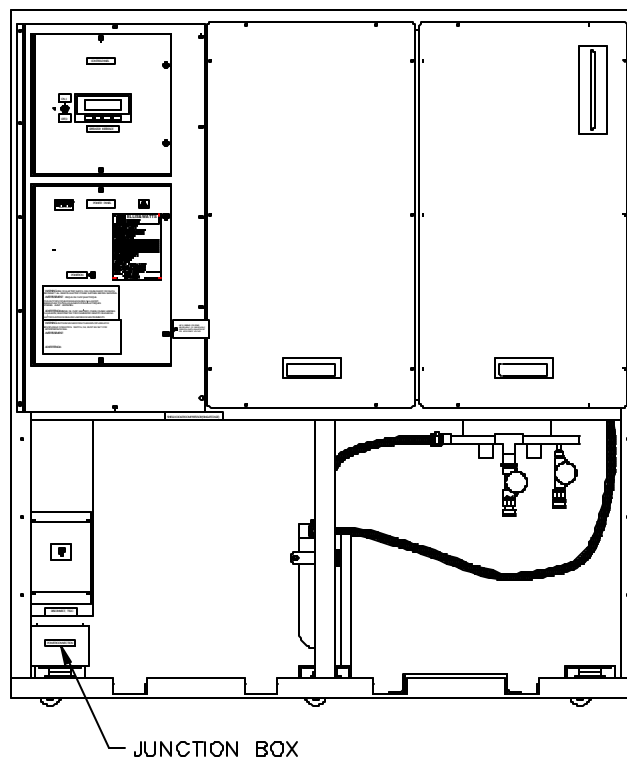


Illustration 20 **JUNCTION BOX LOCATION**

2. Remove the top junction box cover by loosening the four cover screws and sliding the cover off.

NOTE: Ellis & Watts recommends using a minimum conductor size of 6 AWG. Maximum size recommended is 2 AWG. Ellis & Watts recommends using 1 inch (minimum) Liquidtite (or equivalent) conduit and fittings. The installer supplies wire, conduit, and fittings.

3. Route the four power conductors through knockout provided on the junction box.

NOTE: It is very important to maintain correct phasing.

4. Strip the wire and insert into the terminals. When the conductor is properly installed, there should be no bare wire exposed. Pay special attention to the phase rotation.

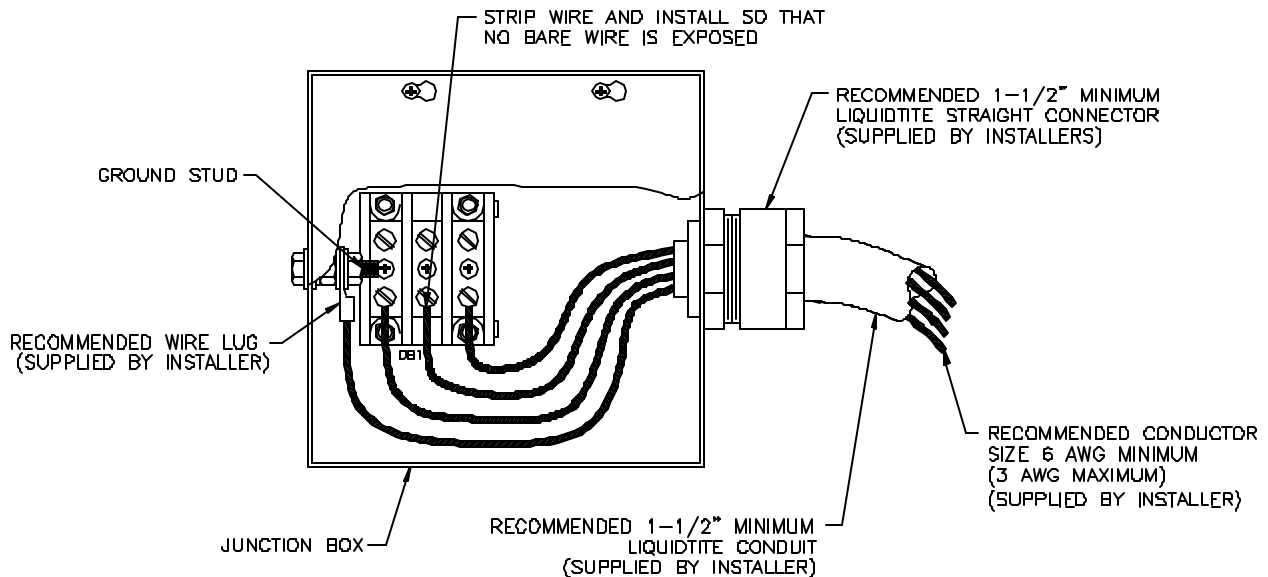


Illustration 21 JUNCTION BOX WIRING

5. Attach the ground wire to the ground stud provided on the junction box.
6. Reinstall the junction box cover.

5-5 ASSEMBLING THE INTERCONNECTING HOSES

1. Verify the parts and quantities from the FI-0035 drawing. Refer to Table 12

Refer to drawing FI-0035 that comes with the ship loose parts, for detailed illustrations.

The following items are provided by the installer and are required for the proper installation of the interconnecting hoses.

- Light oil or soapy water
 - Loctite 567 or equivalent
2. Cut 2 pieces of 1/2 inch tubing to length required (maximum 60ft. (18.3 m)) cleanly and squarely with a razor knife.
 3. Lubricate the barb fitting and/or hose I.D. with a light oil or soapy water only. Do not use heavy oil or grease.

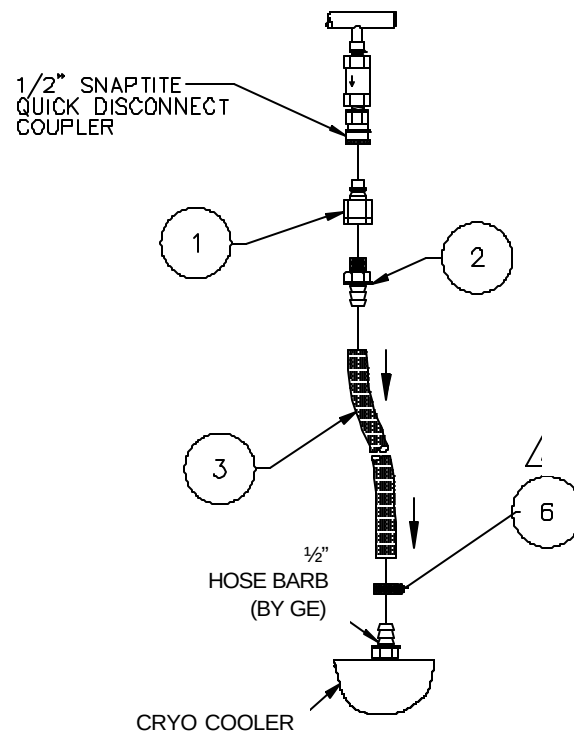


Illustration 22 HOSE TO HOSE BARB ASSEMBLY

Refer to table 11, item numbers 1 to 3, 6

4. Insert the fitting into the hose until the barb is in the hose.

5. Place the end of the fitting against a wood block. Grip the hose firmly at approximately one inch (2.54 cm) from the end of the hose. Push with steady force until the end of the hose bottoms on the fitting and is covered by the yellow plastic cap.
6. Apply a generous amount of LocTite 567 or equivalent to the male pipe threads of each hose barb fitting.

Assemble and mark each hose per table.

HOSE SECTION Feet (m)	QUICK DISCONNECT COUPLER	MARK HOSE AS:
60	Male	Cryo Cooler coolant from TSCC
60	Female	Cryo Cooler coolant to TSCC

Table 16 **HOSE AND QUICK DISCONNECT**

7. Tighten each threaded connection snugly to prevent leaks.
8. Using hose clamps (installer supplied) attach each hose assembly to appropriate connection of Cryo Cooler.

5-6 CRYO COOLER COMPRESSOR INSTALLATION

NOTE: If the installer wants or is able to install the Cryo Cooler Compressor refer to drawing FI-0033 that comes with the ship loose parts, for detailed illustrations.

Refer to Table 16, items 1 and 2.

1. Align the ramps with the two slots located on the right front side of the unit. Hook ramp in place.

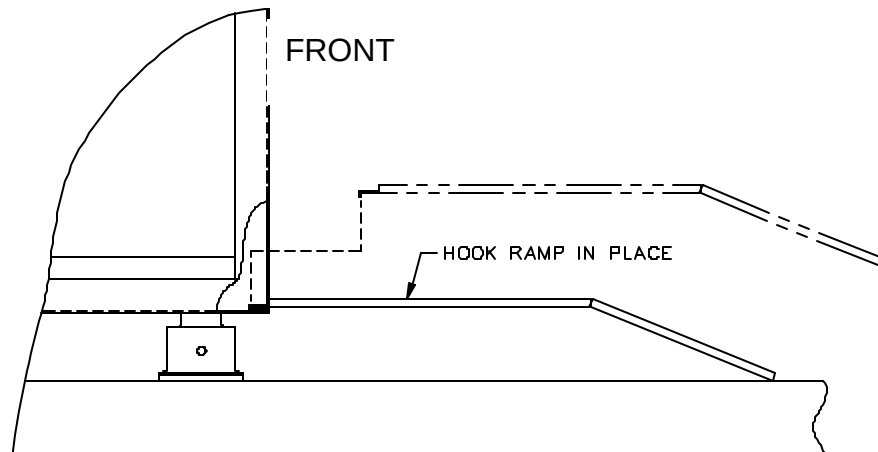


Illustration 23 **RAMP PLACEMENT**

CAUTION

BE CAREFUL WITH ATTACHED HOSES WHEN INSTALLING CRYO COOLER COMPRESSOR.

2. Roll Cryo Cooler compressor in place (see Illustration 24).

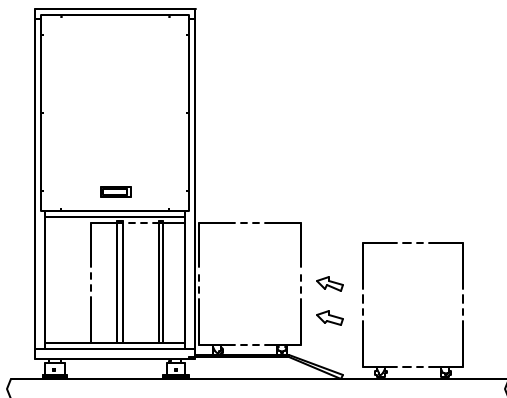


Illustration 24 **CRYO COOLER COMPRESSOR INSTALLATION**

3. Install two strap assemblies with two #10 screws tighten to approximately 24 inches – pounds.
4. Tighten straps as required.

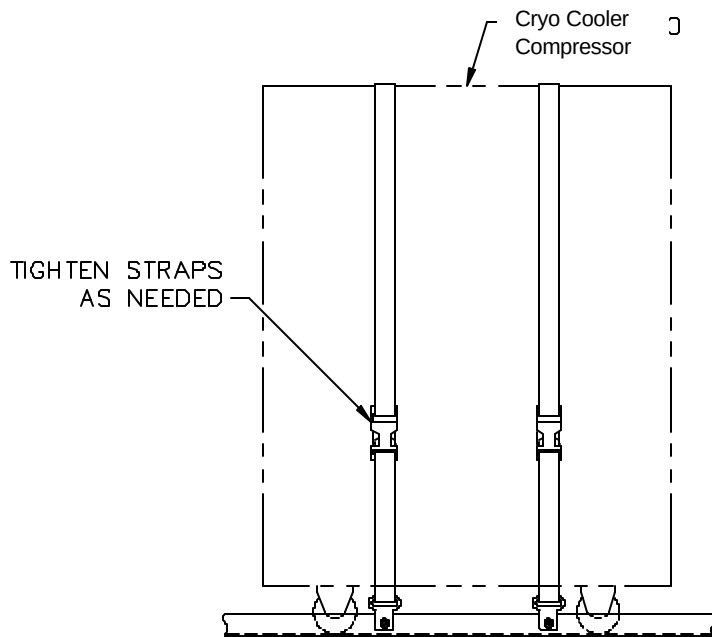


Illustration 25 **SHIELD/CRYO COOLER TIE-DOWN STRAPS**

5. Remove the two ramps and reinstall on the front right side of the unit.

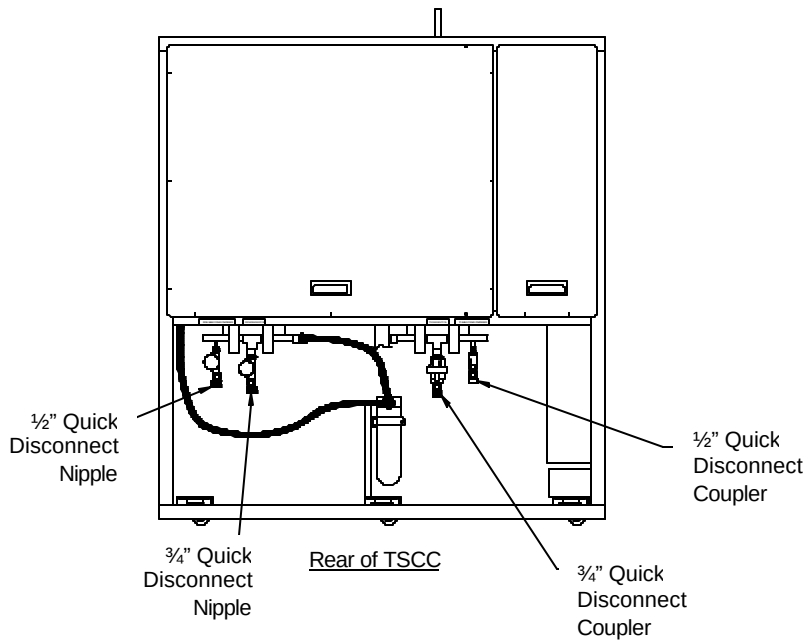


Illustration 26 **HOSE CONNECTION LOCATION**

6 SYSTEM OPERATION

The following procedures are intended to give instruction on the TSCC system controls and routine operation. Initial start-ups after installation, will be performed by GE personnel and system operation will be verified after installation by an Ellis and Watts approved service provider. Start-ups after repairs should be conducted by the customers service provider. It is assumed at this point, that the TSCC and associated equipment have gone through initial startup previously and is ready for operation. If the unit has stopped due to a failure, refer to section 6-1-4.

6-1 OPERATOR START-UP PROCEDURE

6-1-1 Starting Up The TSCC and Associated Equipment

1. Verify that all fluid lines are connected and not leaking. The fluid lines are located on the rear of the unit.
2. Check the coolant level in the reservoir. Verify that the coolant level is between the add and full labels. The reservoir sight glass is located on the top right (See Illustration 18). If it is not refer to Section 6-1-4.

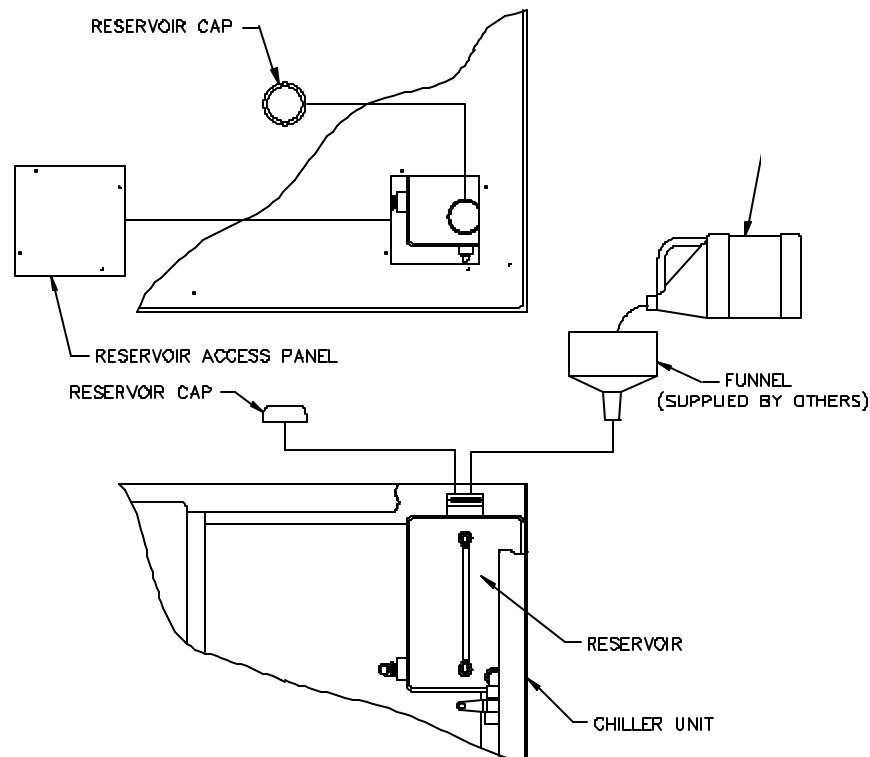


Illustration 27 **RESERVOIR SIGHT GLASS**

CAUTION

During periods where compressor crankcase heaters are without power, liquid refrigerant can accumulate in compressor crankcase. Restarting compressor without allowing sufficient time for crankcase heaters to vaporize liquid refrigerant can cause serious internal damage to compressor. Wait a minimum of two hours after applying power.

3. The Control Power Switch is located on the top front panel (see Illustration 28). Verify toggle switch is switched off. If the main disconnect switch is OFF, set to ON by rotating red handle clockwise until handle is in vertical position. The main disconnect switch is located on the front lower left side of the unit (see Illustration 18).
4. Place the Control Power Switch to the ON Position. TSCC should start after 5 seconds and be fully functional after 30 seconds. If the TSCC shuts down after 30 seconds, refer to section 6-1-4.

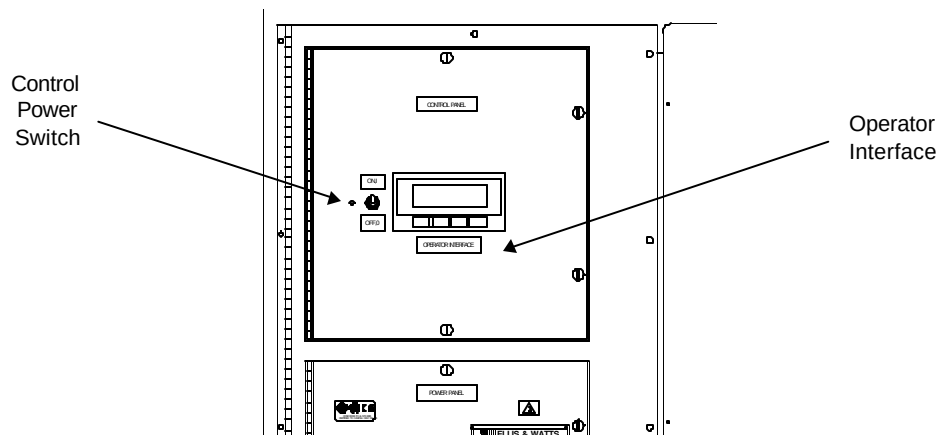


Illustration 28 **CONTROL PANEL**

After 30 minutes of operation verify that the unit is maintaining proper temperatures and flows as shown in table 17.

NOTE: Two globe valves are located on the coolant supply lines (gradient and cryo loops) for flow adjustment. The valves have been set for 60 Hz operation. For 50 Hz operation, the valves should be adjusted to obtain the flow shown in Table 17.

	Coolant Flow	Coolant Temperature
Gradient	6.0 GPM + 0.5 GPM	20° +/- 1C
Cryo Cooler Flow	2.1 GPM + 0.5 GPM	20° +/- 1C

Table 17 **TSCC TEMPERATURE AND FLOW TOLERANCES**

6-1-2 Shutting Down The TSCC

1. Verify that the red led on the Operator Interface is not on. If the led is glowing, check the alarm section of the Operator Interface for any faults and log them before proceeding. Refer to section 6-1-4 for further instructions.
2. Set the Control Power Switch to OFF.

6-1-3 Maintenance

An approved service provider should conduct routine maintenance every 4 months. This maintenance consists of:

Verifying proper flows and temperatures per table 13 and logging data.

Change coolant filter.

Leak test refrigerant system.

Check electrical control panel for burned or frayed wiring and loose connections.

Checking all coolant hoses for obvious leaks and for worn areas.

Verifying proper operation of refrigeration system.

Check condenser coils for dirt or damage.

Contact GE for current schedule or approved service provider.

6-1-4 General Troubleshooting

The TSCC chiller system is designed to be monitored by the GE Service Center at all times. Some occasions arise where the operator can assist in troubleshooting. In all cases the operator will be **most useful** by checking for and logging any faults on the Operator Interface and noting any abnormal conditions in the equipment room where the TSCC is located. Refer to sections 6-2-1 through 6-2-3 for instruction on how to locate faults on Operator Interface screens.

6-2 PLC OPERATION

6-2-1 Operator Interface Display Description:

The operator interface is located on the front panel of the TSCC and displays the model number of the unit, the software revision of the programmed PLC, parameter set points, parameter data, parameter alarm status, and system run time. The interface also allows the user to change the operating set points for the fluid coolant circuit. The upper and lower alarm limits for the various parameters are preset by the manufacturer and are not accessible by service. See Illustration 29 for an example of the layout of the display, user buttons, power switch and indicator lights.

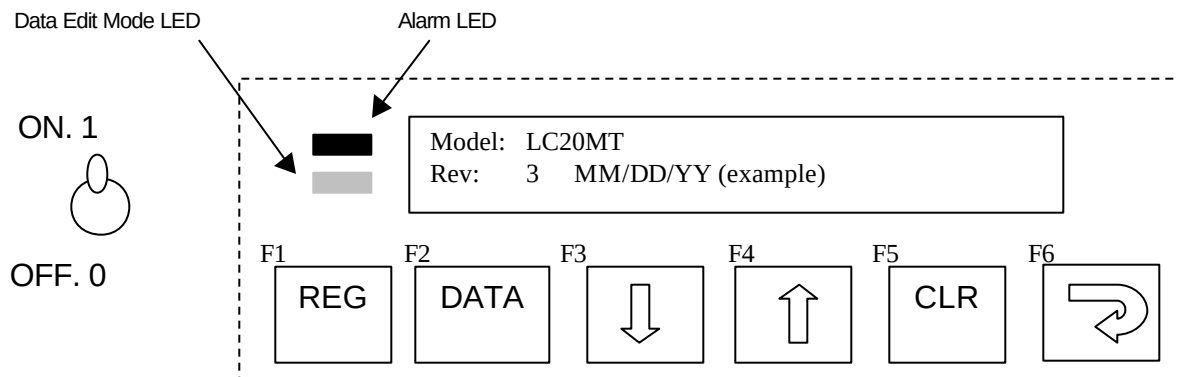


Illustration 29 **OPERATOR INTERFACE**

6-2-2 Button Description

- Button F1 Toggles the display between Menu 1 and Menu 2.
- Button F2 Puts the unit into data edit mode for changing the operating set point for supply coolant temperature. The green led will light when the unit is in data edit mode.
- Button F3 Menu List Display: Steps through the menu list in the forward direction. Change Set Point: Increases the value shown on the display in 0.1 unit increments.
- Button F4 Menu List Display: Steps through the menu list in the reverse direction. Change Set Point: Decreases the value shown on the display in 0.1 unit increments.

- Button F5 To extinguish the red alarm status indicator. Note: The operating parameter must be in a normal or no fault mode in order to turn off the red alarm status indicator.
- Button F6 Stores new set point changes and exits the data edit mode.

6-2-3 Menu Lists

1. Press the F1 Button to change between Menu 1 and Menu 2.
2. Refer to Table 19, Menu 1 – System Parameter Data, to view the listing of the following types of items: Model Number and Software revision of the unit, operating set points for specific parameters, current parameter data measurement, and total run hours. The first line item on Menu 1 will show the Model Number of the Unit.
3. Refer to Table 20, Menu 2 – System Operating Status, to view the listing of the alarm or failure conditions. The first line item on Menu 2 will show “Supply Fluid Temperature”.

6-2-4 Changing The Set Points

The coolant temperature set points are preset at Ellis and Watts per G.E. specifications. There is a 2°C allowable variance, settable by the operator. If there is a need to change the set point value, then refer to the following steps to perform the change.

1. Press F3 to step forward through the menu until the display shows the coolant temp set point.
2. Press F2 to enter the data edit mode.
3. Press either the F3 or F4 buttons to change the set point value.
4. Press F6 to store the set point values and exit the data edit mode.

Menu 1 – System Parameter Data

Example of Model Number and Revision	
Model Number:	LC20MT
Rev.: #	Mo/Dy/Yr
Parameter	Range Value / Unit of Measure (see Note 1)
Supply Temp:	0 to 29.4 Degrees Centigrade
Return Temp:	0 to 50 Degrees Centigrade
Temp Set Point:	17 to 24 Degrees Centigrade (Factory set 20°C)
Gradient Flow:	0 to 6.0 GPM +/- 10%
Cryo-Cooler Flow:	0 to 2.5 GPM +/- 10%
TSCC 3 way valve % open:	0 to 100%
Total Run Hours:	0 to 9999 Hours
NOTE 1: Unit of measure is set at the factory.	
NOTE 2: After the run hours reaches 9999, the unit resets to zero.	

Table 18 **SYSTEM PARAMETER DATA****Menu 2 – System Operating Status**

Parameter	Display of Operating Mode		
Supply Temp:	Low	Normal	High
Coolant Level:	Low	Normal	
Gradient Flow:	Failure	Low	Normal
Cryo-Cooler Flow:	Failure	Low	Normal
Suction Pressure	Fault	No Fault	
Discharge Pressure	Fault	No Fault	
Compressor / Phase	Fault	No Fault	
Compressor Cycling:	Normal	- - - - (See Note 1)	
NOTE 1: Each dashed line represents an attempt by the PLC to restart the chiller. After 5 attempts the TSCC will shut down.			

Table 19 **SYSTEM OPERATING STATUS**

TERMS & ACRONYMS

GEMS	General Electric Medical Systems
MDP	Main Distribution Panel
QD	Quick disconnect coupler
RGA	Returned Goods Authorization
EPA	Environmental Protection Agency
MRI	Magnetic Resonance Imaging
UBC	United Builders Code
SST	Stainless Steel
NPTF	National Pipe Thread, Female
DB	Distribution Block
PLC	Programmable Logic Control
TSCC	Twin System Cooling Cabinet
Liquidtite	Trade name for water tight flexible conduit
LocTite	Brand name for adhesives, sealants, and threadlockers

INDEX OF TABLES

Table 1	INSTALLATION TOOLS AND SUPPLIES	6
Table 2	INSTALLATION RESPONSIBILITY MATRIX	7
Table 3	SHIPPING DIMENSIONS	12
Table 4	ENVIRONMENTAL CONDITIONS	13
Table 5	OPERATING SPECIFICATIONS	15
Table 6	POWER REQUIREMENTS	15
Table 7	UNCRATED DIMENSIONS	16
Table 8	DOOR/HALLWAY SIZES	18
Table 9	FLOOR LOADING REQUIREMENTS	20
Table 10	SEISMIC TIE DOWN KIT	28
Table 11	PIPING/HOSING CONNECTIONS KIT	28
Table 12	OTHER DRAWINGS	28
Table 13	AIR DAM	29
Table 14	SPARE PARTS	29
Table 15	SHIELD COOLER INSTALLATION	29
Table 16	HOSE AND QUICK DISCONNECT	40
Table 17	TSCC TEMPERATURE AND FLOW TOLERANCES	45
Table 18	SYSTEM PARAMETER DATA	49
Table 19	SYSTEM OPERATING STATUS	49

INDEX OF ILLUSTRATIONS

Illustration 1	PLACEMENT OF TSCC TO MAGNET	14
Illustration 2	NECESSARY CLEARANCES FOR TSCC	16
Illustration 3	TSCC OVERALL DIMENSIONS	17
Illustration 4	MINIMUM HALLWAY DIMENSION FOR TURNS ([] = cm)	18
Illustration 5	MINIMUM DOORWAY DIMENSIONS ([] = cm)	19
Illustration 6	VIBRATION ISOLATOR LAYOUT DIMENSIONS	21
Illustration 7	TSCC TIE DOWN BRACKET PLACEMENT	22
Illustration 8	INTERNATIONAL CRATE EXAMPLE	23
Illustration 9	UNCRATING	24
Illustration 10	TSCC LIFTING METHOD	25
Illustration 11	C-CLAMP PLACEMENT	26
Illustration 12	ACCEPTABLE ALTERNATIVE LIFTING METHOD	27
Illustration 13	UNACCEPTABLE ALTERNATIVE LIFTING METHOD	27
Illustration 14	ISOLATOR LAYOUT PATTERN	30
Illustration 15	VIBRATION ISOLATOR MOUNTING	31
Illustration 16	TIE-DOWN BRACKET LAYOUT PATTERN	33
Illustration 17	½ Inch Bolt Assembly	34
Illustration 18	TIE-DOWN BRACKET MOUNTING	35
Illustration 19	AIR DAM INSTALLATION	36
Illustration 20	JUNCTION BOX LOCATION	37
Illustration 21	JUNCTION BOX WIRING	38
Illustration 22	HOSE TO HOSE BARB ASSEMBLY	39
Illustration 23	RAMP PLACEMENT	41
Illustration 24	CRYO COOLER COMPRESSOR INSTALLATION	41
Illustration 25	SHIELD/CRYO COOLER TIE-DOWN STRAPS	42
Illustration 26	HOSE CONNECTION LOCATION	43
Illustration 27	RESERVOIR SIGHT GLASS	44
Illustration 28	CONTROL PANEL	45
Illustration 29	OPERATOR INTERFACE	47

MATERIAL SAFETY DATA SHEETS

Forane® R - 134a
Tetrafluoroethane (HFC - 134a)

Dowfrost® Heat Transfer Fluid
Propylene glycol
Dipotassium Phosphate
Deionized Water

1. CHEMICAL PRODUCT & COMPANY IDENTIFICATION **Page: 1**

24-Hour Emergency Phone Number: 989-636-4400

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: 01/08/99 Date Printed: 05/07/01 MSD: 000130

The Dow Chemical Company, Midland, MI 48674

Customer Information Center: 800-258-2436

2. COMPOSITION/INFORMATION ON INGREDIENTS

Propylene glycol	CAS# 000057-55-6	95%
Dipotassium phosphate	CAS# 007758-11-4	<5%
Deionized water	CAS# 007732-18-5	<5%

3. HAZARDS IDENTIFICATION

EMERGENCY OVERVIEW

* Colorless, odorless liquid. Toxic fumes released in fire * situations.

POTENTIAL HEALTH EFFECTS (See Section 11 for toxicological data.)

EYE: May cause slight transient eye irritation. Corneal injury is unlikely.

SKIN: Essentially nonirritating to skin on prolonged contact. A single prolonged skin exposure is not likely to result in the material being absorbed through skin in harmful amounts. Repeated exposures may cause slight flaking, tenderness and softening of skin.

(Continued on page 2)

OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

Product: DOWFROST* HEAT TRANSFER FLUID Product Code: 23545

Effective Date: 01/08/99 Date Printed: 05/07/01 MSD: 000130

INGESTION: Single dose oral toxicity is low. No hazards anticipated from swallowing small amounts incidental to normal handling operations.

INHALATION: A single prolonged (hours) inhalation exposure is not likely to cause adverse effects. Mists are not likely to be hazardous.

SYSTEMIC (OTHER TARGET ORGAN) EFFECTS: Repeated excessive ingestion may cause central nervous system effects.

CANCER INFORMATION: Did not cause cancer in long-term animal studies.

TERATOLOGY (BIRTH DEFECTS): Birth defects are unlikely. Exposures having no adverse effects on the mother should have no effect on the fetus.

REPRODUCTIVE EFFECTS: In animal studies, has been shown not to interfere with reproduction.

4. FIRST AID

EYE: Flush eyes with plenty of water.

SKIN: Wash off in flowing water or shower.

INGESTION: If swallowed, seek medical attention. Do not induce vomiting unless directed to do so by medical personnel.

INHALATION: Remove to fresh air if effects occur. Consult a physician.

NOTE TO PHYSICIAN: No specific antidote. Supportive care. Treatment based on judgment of the physician in response to reactions of the patient.

5. FIRE FIGHTING MEASURES

FLASH POINT: 215F, 102C METHOD USED: TOC (Tag Open Cup)

(Continued on page 3)

* OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: 01/08/99 Date Printed: 05/07/01 MSD: 000130

FLAMMABLE LIMITS LFL: 2.6% @ 100C UFL: 12.5% @ 130C

HAZARDOUS COMBUSTION PRODUCTS: During a fire, smoke may contain the original material in addition to unidentified toxic and/or irritating compounds. Hazardous combustion products may include and are not limited to carbon monoxide, carbon dioxide.

OTHER FLAMMABILITY INFORMATION: Violent steam generation or eruption may occur upon application of direct water stream. Flammable concentration of vapor can accumulate at temperatures above 215.0 deg. F. Liquid mist of this product can burn. Spills of these organic liquids on hot fibrous insulations may lead to lowering of the autoignition temperatures possibly resulting in spontaneous combustion.

EXTINGUISHING MEDIA: Water fog or fine spray, carbon dioxide, dry chemical, foam. Alcohol resistant foams (ATC type) are preferred if available. General purpose synthetic foams (including AFFF) or protein foams may function, but much less effectively. Do not use direct water stream. Will spread fire.

MEDIA TO BE AVOIDED: Do not use direct water stream.

FIRE FIGHTING INSTRUCTIONS: Keep people away. Isolate fire area and deny unnecessary entry. Burning liquids may be moved by flushing with water to protect personnel and minimize property damage. Burning liquids may be extinguished by dilution with water. Do not use direct water stream. May spread fire.

PROTECTIVE EQUIPMENT FOR FIRE FIGHTERS: Wear positive-pressure self-contained breathing apparatus (SCBA) and protective firefighting clothing (includes fire-fighting helmet, coat, pants, boots, and gloves). If protective equipment is not available or not used, fight fire from a protected location or safe distance.

6. ACCIDENTAL RELEASE MEASURES (See Section 15 for Regulatory Information)

(Continued on page 4)

* OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: 01/08/99 Date Printed: 05/07/01 MSD: 000130

PROTECT PEOPLE: Clear non-emergency personnel from area.

PROTECT THE ENVIRONMENT: Contain liquid to prevent contamination of soil, surface water or ground water.

CLEANUP: Clean up with absorbent material. Sweep up.

7. HANDLING AND STORAGE

HANDLING: Product on surfaces can cause slippery conditions.

STORAGE: Store below 121 C, 250 F.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

ENGINEERING CONTROLS: Good general ventilation should be sufficient for most conditions.

PERSONAL PROTECTIVE EQUIPMENT

EYE/FACE PROTECTION: Use safety glasses.

SKIN PROTECTION: For brief contact, no precautions other than clean body-covering clothing should be needed. Use impervious gloves when prolonged or frequently repeated contact could occur.

RESPIRATORY PROTECTION: No respiratory protection should be needed.

EXPOSURE GUIDELINE(S): Propylene glycol: AIHA WEEL is 50 ppm total, 10 mg/m3 aerosol only.

9. PHYSICAL AND CHEMICAL PROPERTIES

APPEARANCE: Colorless. ODOR: Odorless liquid. VAPOR PRESSURE: 0.22 mmHg @ 20C, 68F VAPOR DENSITY: 2.62 BOILING POINT: 370F, 188C SOLUBILITY IN WATER: Complete

(Continued on page 5)

* OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: 01/08/99 Date Printed: 05/07/01 MSD: 000130

SPECIFIC GRAVITY: 1.050 @ 60/60F, 16C

10. STABILITY AND REACTIVITY

CHEMICAL STABILITY: Thermally stable at typical use temperatures.

CONDITIONS TO AVOID: Avoid temperatures above 121C/250F. Product can decompose at elevated temperatures.

INCOMPATIBILITY WITH OTHER MATERIALS: Avoid contact with oxidizing materials.

HAZARDOUS DECOMPOSITION PRODUCTS: Hazardous decomposition products depend upon temperature, air supply and the presence of other materials.

HAZARDOUS POLYMERIZATION: Will not occur.

11. TOXICOLOGICAL INFORMATION (See Section 3 for Potential Health Effects. For detailed toxicological data, write or call the address or non-emergency number shown in Section 1)

SKIN: The LD50 for skin absorption in rabbits is greater than 10,000 mg/kg.

INGESTION: The oral LD50 for female rats is about 20.3 g/kg.

MUTAGENICITY (EFFECTS ON GENETIC MATERIAL): Results of in vitro (test tube) mutagenicity tests have been negative. Results of mutagenicity tests in animals have been negative.

12. ECOLOGICAL INFORMATION (For detailed Ecological data, write or call the address or non-emergency number shown in Section 1)

ENVIRONMENTAL FATE MOVEMENT AND PARTITIONING: Based largely or completely on data for major component(s). Bioconcentration potential is low (BCF less than 100 or Log Pow less than 3).

DEGRADATION AND PERSISTENCE: Based largely or completely on data for major component(s). Biodegradation under aerobic static laboratory conditions is high (BOD20 or BOD28/ThOD

(Continued on page 6)

* OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: 01/08/99 Date Printed: 05/07/01 MSD: 000130

greater than 40%).

ECOTOXICITY: Based largely or completely on data for major component(s). Material is practically non-toxic to aquatic organisms on an acute basis (LC50 greater than 100 mg/L in most sensitive species).

13. DISPOSAL CONSIDERATIONS (See Section 15 for Regulatory Information)

DISPOSAL: DO NOT DUMP INTO ANY SEWERS, ON THE GROUND OR INTO ANY BODY OF WATER. All disposal methods must be in compliance with all Federal, State/Provincial and local laws and regulations. Regulations may vary in different locations. Waste characterizations and compliance with applicable laws are the responsibility solely of the waste generator. THE DOW CHEMICAL COMPANY HAS NO CONTROL OVER THE MANAGEMENT PRACTICES OR MANUFACTURING PROCESSES OF PARTIES HANDLING OR USING THIS MATERIAL. THE INFORMATION PRESENTED HERE PERTAINS ONLY TO THE PRODUCT AS SHIPPED IN ITS INTENDED CONDITION AS DESCRIBED IN MSDS SECTION 2 (Composition/information On Ingredients).

FOR UNUSED & UNCONTAMINATED PRODUCT, the preferred options include sending to a licensed, permitted: recycler, reclaimer, or waste water treatment system.

As a service to its customers, Dow can provide names of information resources to help identify waste management companies and other facilities which recycle, reprocess or manage chemicals or plastics, and that manage used drums. Telephone Dow's Customer Information Center at 800-258-2436 or 517-832-1556 for further details.

14. TRANSPORT INFORMATION

CANADIAN TDG INFORMATION: For TDG regulatory information, if required, consult transportation regulations, product shipping papers, or your Dow representative.

15. REGULATORY INFORMATION (Not meant to be all-inclusive- -selected regulations represented)

NOTICE: The information herein is presented in good faith and

(Continued on page 7)

* OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: 01/08/99 Date Printed: 05/07/01 MSD: 000130

believed to be accurate as of the effective date shown above. However, no warranty, express or implied is given. Regulatory requirements are subject to change and may differ from one location to another; it is the buyer's responsibility to ensure that its activities comply with federal, state or provincial, and local laws. The following specific information is made for the purpose of complying with numerous federal, state or provincial, and local laws and regulations. See other sections for health and safety information.

U.S. REGULATIONS

SARA 313 INFORMATION: To the best of our knowledge, this product contains no chemical subject to SARA Title III Section 313 supplier notification requirements.

SARA HAZARD CATEGORY: This product has been reviewed according to the

EPA "Hazard Categories" promulgated under Sections 311 and 312 of the Superfund Amendment and Reauthorization Act of 1986 (SARA Title 111) and is considered, under applicable definitions, to meet the following categories:

Not to have met any hazard category

TOXIC SUBSTANCES CONTROL ACT (TSCA):

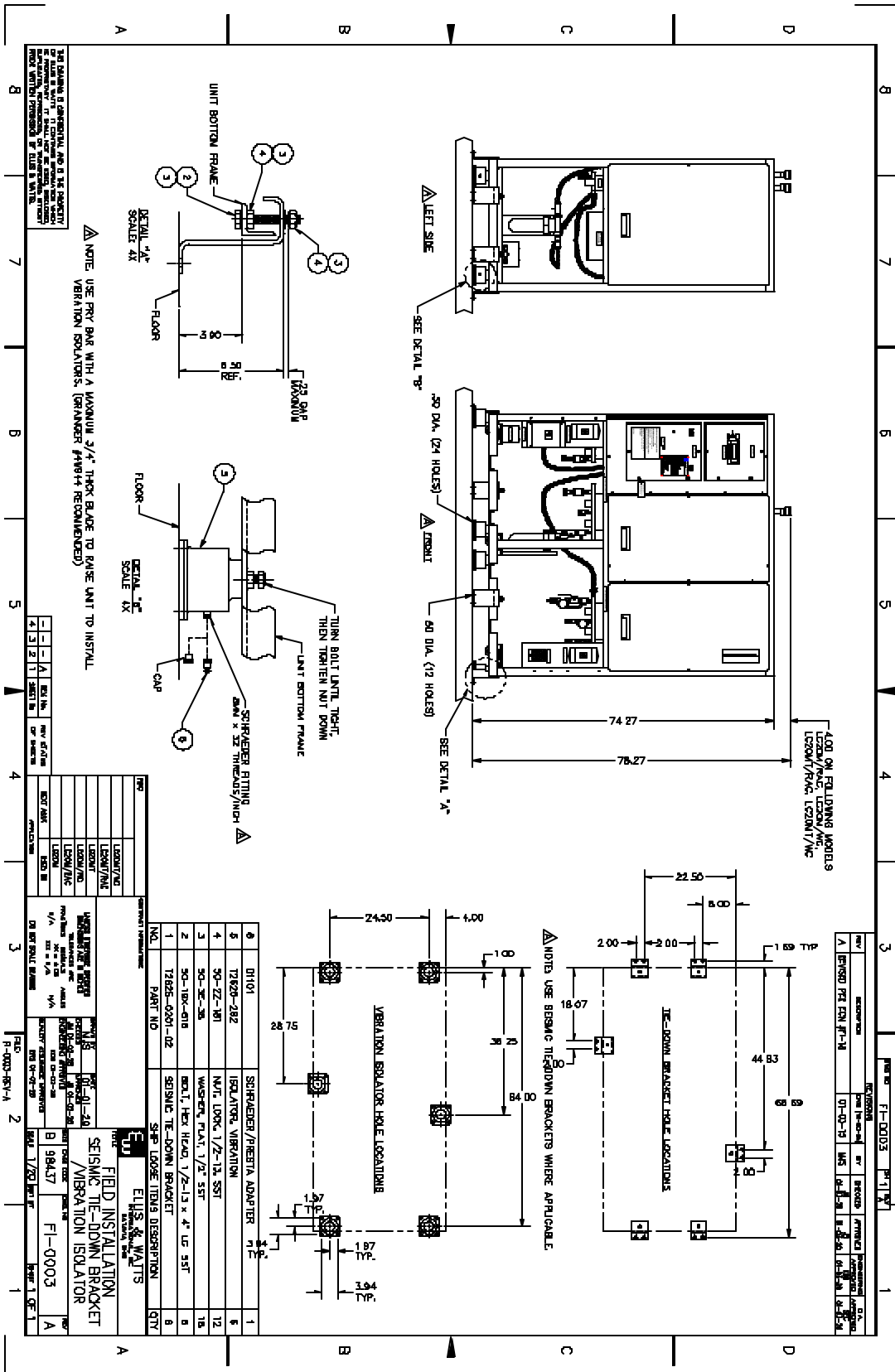
All ingredients are on the TSCA inventory or are not required to be listed on the TSCA inventory.

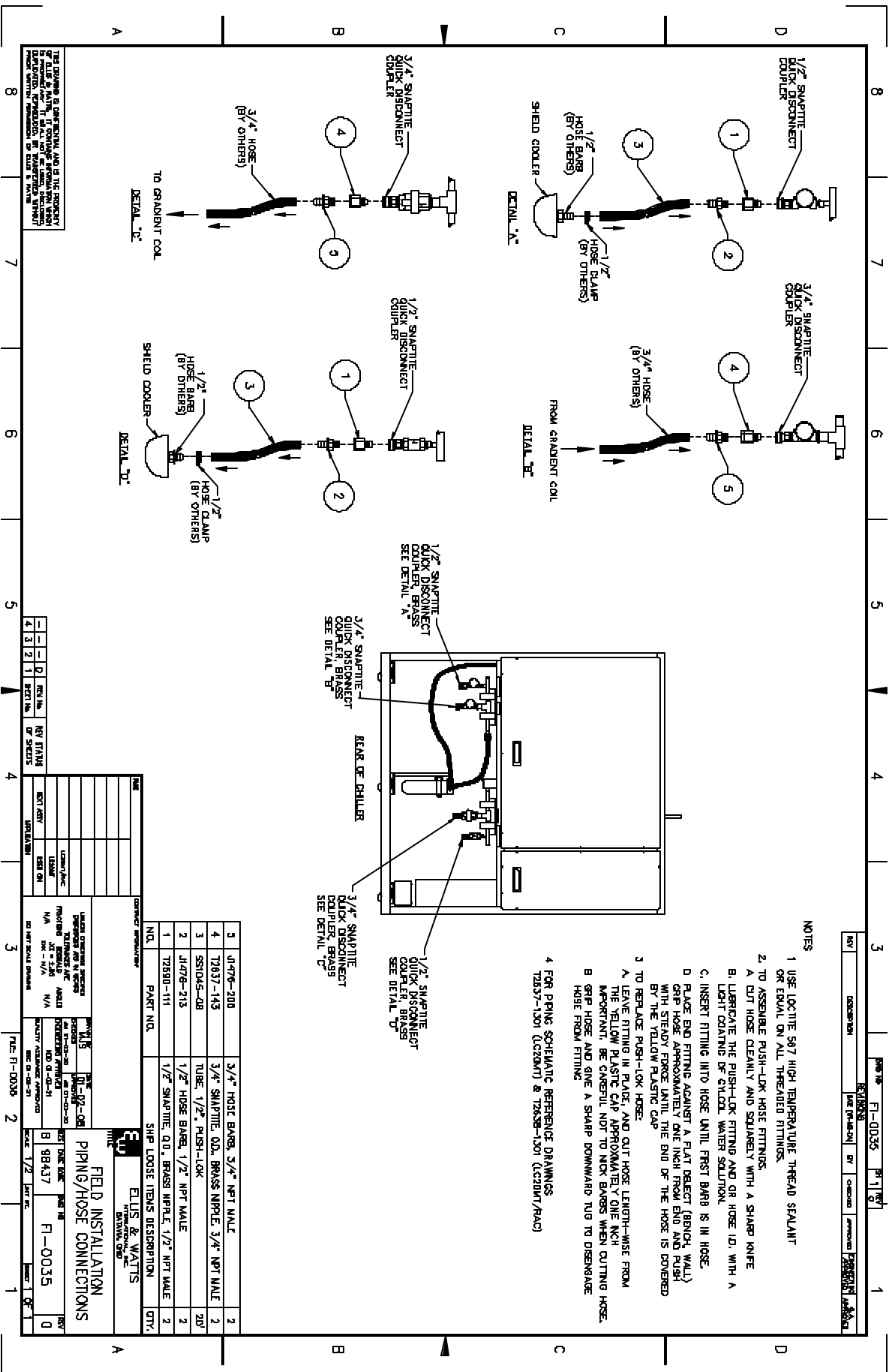
STATE RIGHT-TO-KNOW: The following product components are cited on certain state lists as mentioned. Non-listed components may be shown in the composition section of the MSDS.

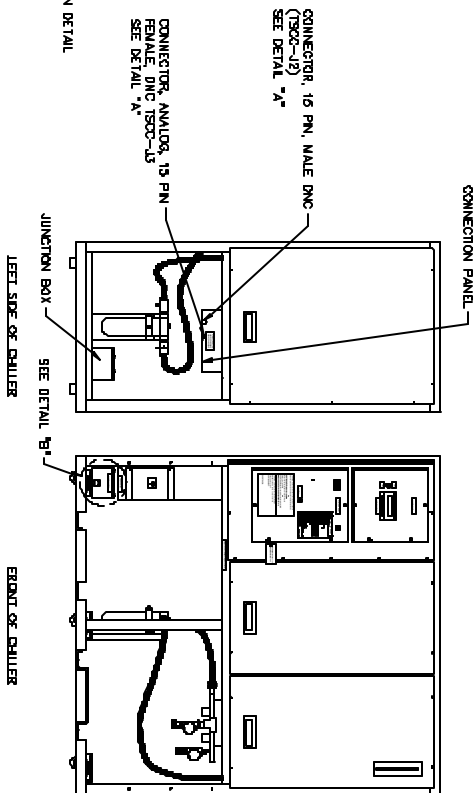
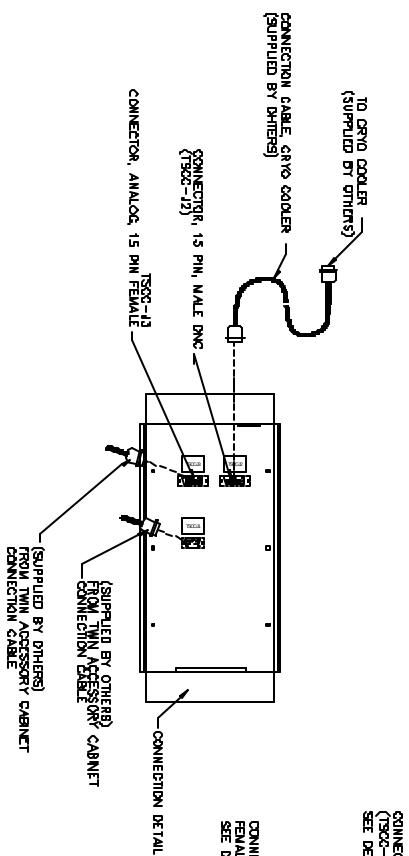
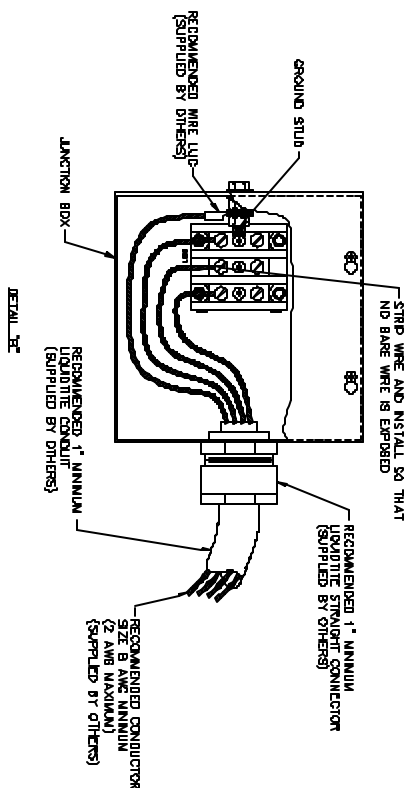
(Continued on page 8)

* OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

FI DRAWINGS







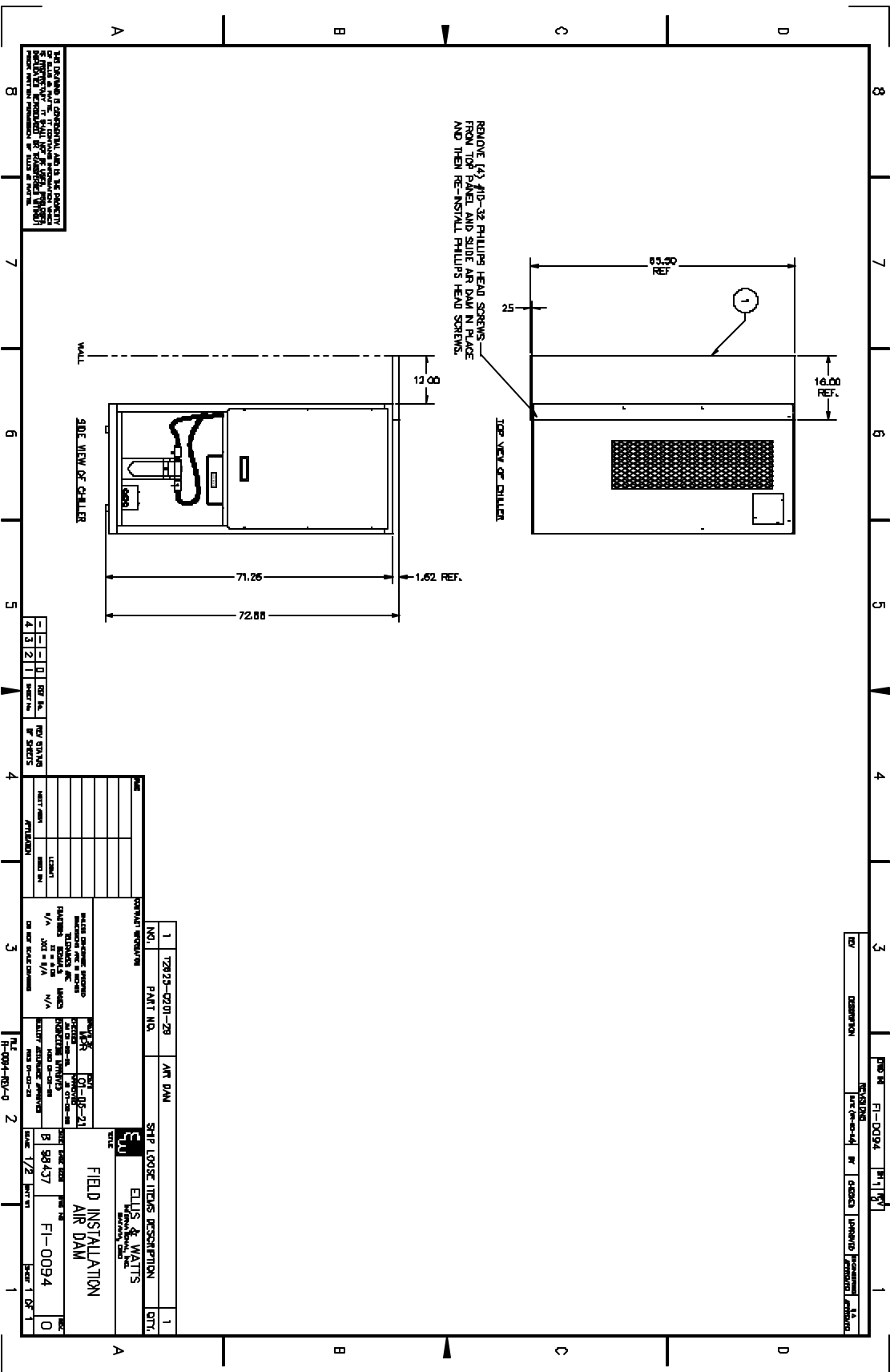
REVISIONS									
REV	DESCRIPTION	DATE (Y-M-D)	BY	REASON	APPROVED	DISAPPROVED	DATE	BY	REASON
A	REMOVED FOR ECH F-44	01-05-21	WPR	REMOVED FOR ECH F-44	REMOVED FOR ECH F-44				
B	REMOVED FOR ECH F-71	01-06-11	MIS	REMOVED FOR ECH F-71	REMOVED FOR ECH F-71				
C	REMOVED FOR ECH F-123	01-07-11	WPR	REMOVED FOR ECH F-123	REMOVED FOR ECH F-123				

THIS ADVERTISING REPRESENTATION IS THE PROPERTY OF ELLIS & FLATTS. IT CANNOT BE REPRODUCED FOR ANY OTHER PURPOSE WITHOUT THE WRITTEN PERMISSION OF ELLIS & FLATTS.

-	-	-	C	REV No	REV STATUS OF SHEETS
4	3	2	1	SHEET No.	

 ELIUS & WATTS ELECTRICAL CONTRACTORS		FIELD INSTALLATION ELECTRICAL / POWER / COMMUNICATION CONNECTIONS		DATE: _____ TIME: _____
PROJECT NO: _____ PROJECT NAME: _____ PROJECT ADDRESS: _____ PROJECT CITY: _____ PROJECT STATE: _____ PROJECT ZIP: _____		CLIENT NAME: _____ CLIENT ADDRESS: _____ CLIENT CITY: _____ CLIENT STATE: _____ CLIENT ZIP: _____		
PROJECT DESCRIPTION: _____ PROJECT SCOPE: _____ PROJECT START DATE: _____ PROJECT END DATE: _____		PROJECT STATUS: _____ PROJECT PHASE: _____ PROJECT BUDGET: _____ PROJECT COST: _____		
PROJECT MANAGER: _____ PROJECT SUPERVISOR: _____ PROJECT ENGINEER: _____ PROJECT DESIGNER: _____		PROJECT SCHEDULE: _____ PROJECT Gantt Chart: _____ PROJECT RISK: _____ PROJECT QUALITY: _____		
PROJECT CONTACT: _____ PROJECT PHONE: _____ PROJECT FAX: _____ PROJECT EMAIL: _____		PROJECT WEBSITE: _____ PROJECT SOCIAL MEDIA: _____ PROJECT BLOG: _____ PROJECT NEWS: _____		

NOTE FOR ELECTRICAL SCHEMATIC REFERENCE DRAWING T2637-1204



THE DRAWING IS CONSIDERED A PART OF THE PROJECT AND IS NOT TO BE REPRODUCED OR USED IN ANY MANNER WITHOUT THE WRITTEN PERMISSION OF THE PROJECT MANAGER.

1	2	3	4	5	6	7	8
1	2	3	4	5	6	7	8

REV	DESCRIPTION	DATE	BY	CHKD
1	12922-0210-208	12-05-21	ELLS & WATTS	
2	12922-0210-208	12-05-21	ELLS & WATTS	
3	12922-0210-208	12-05-21	ELLS & WATTS	
4	12922-0210-208	12-05-21	ELLS & WATTS	
5	12922-0210-208	12-05-21	ELLS & WATTS	
6	12922-0210-208	12-05-21	ELLS & WATTS	
7	12922-0210-208	12-05-21	ELLS & WATTS	
8	12922-0210-208	12-05-21	ELLS & WATTS	
9	12922-0210-208	12-05-21	ELLS & WATTS	
10	12922-0210-208	12-05-21	ELLS & WATTS	

REV	DESCRIPTION	DATE	BY	CHKD
1	12922-0210-208	12-05-21	ELLS & WATTS	
2	12922-0210-208	12-05-21	ELLS & WATTS	
3	12922-0210-208	12-05-21	ELLS & WATTS	
4	12922-0210-208	12-05-21	ELLS & WATTS	
5	12922-0210-208	12-05-21	ELLS & WATTS	
6	12922-0210-208	12-05-21	ELLS & WATTS	
7	12922-0210-208	12-05-21	ELLS & WATTS	
8	12922-0210-208	12-05-21	ELLS & WATTS	
9	12922-0210-208	12-05-21	ELLS & WATTS	
10	12922-0210-208	12-05-21	ELLS & WATTS	

REV	DESCRIPTION	DATE	BY	CHKD
1	12922-0210-208	12-05-21	ELLS & WATTS	
2	12922-0210-208	12-05-21	ELLS & WATTS	
3	12922-0210-208	12-05-21	ELLS & WATTS	
4	12922-0210-208	12-05-21	ELLS & WATTS	
5	12922-0210-208	12-05-21	ELLS & WATTS	
6	12922-0210-208	12-05-21	ELLS & WATTS	
7	12922-0210-208	12-05-21	ELLS & WATTS	
8	12922-0210-208	12-05-21	ELLS & WATTS	
9	12922-0210-208	12-05-21	ELLS & WATTS	
10	12922-0210-208	12-05-21	ELLS & WATTS	

REV	DESCRIPTION	DATE	BY	CHKD
1	12922-0210-208	12-05-21	ELLS & WATTS	
2	12922-0210-208	12-05-21	ELLS & WATTS	
3	12922-0210-208	12-05-21	ELLS & WATTS	
4	12922-0210-208	12-05-21	ELLS & WATTS	
5	12922-0210-208	12-05-21	ELLS & WATTS	
6	12922-0210-208	12-05-21	ELLS & WATTS	
7	12922-0210-208	12-05-21	ELLS & WATTS	
8	12922-0210-208	12-05-21	ELLS & WATTS	
9	12922-0210-208	12-05-21	ELLS & WATTS	
10	12922-0210-208	12-05-21	ELLS & WATTS	